

Architecture Design Document (ADD) v2.7

EAI Integration Roadmap

Version: Version: 2.7

Date: 08 May 2026,

Prepared by: O.C.

Jira Space: EAI Integration Roadmap

Status: (Final Consolidated)

Revision History

Version	Date	Author	Changes
2.4	08 May 2026	O.C.	Major security update – Finalized Claude Security Testing framework for 2-Sprint Production Release.
2.5	08 May 2026	O.C.	Tines replacement for n8n; new Tines Integration Skill (Section 2.7); new stories under AI-122 (Boomi); full GPU section revision (4.3.3) with cost column, simplified headers, Blackwell vs Hopper comparison, NVIDIA Cloud + DGX Cloud options.
2.6	08 May 2026	O.C.	Final full version. All v2.4 baseline + v2.5 updates integrated. Expanded RACI matrix examples, ARCI framework details, ITIL governance exploration, Boomi integration examples, iPaaS alternatives, refined acceptance criteria phrasing, revised RACI matrix formatting, and implementation examples.
2.7	14 May 2026	JC	Revised story numbers to match JIRA board

Table of Contents

1. Introduction
2. Business and Technical Context
3. Architecture Overview
4. Infrastructure and MCP Platform
5. Detailed Phase Architecture (Phases 1–12)

6. Data Flows, Integration Patterns, and Interfaces
7. Quality Attributes and Non-Functional Requirements
8. Project Execution Plans
9. Visual Models and Diagrams
10. Risks, Dependencies, and Mitigation Strategies
11. Appendices
 - Exhibit Q&A – Shared S1 Explanation (standalone exhibit)
 - Appendix 13.1: GitHub Integration Patterns & Authentication (new in v1.11)
 - **Appendix 13.8: MITRE ATLAS Techniques (Expanded) – Claude Security Testing** (*new in v2.4*)
 - **Appendix 13.15: OWASP LLM Risk Mitigations (Expanded) – Claude Security Testing** (*new in v2.4*)
 - **Appendix 13.16: Red-Team Attack Scenarios – Claude Security Testing** (*new in v2.4*)

1. Introduction

1.1 Purpose and Scope of the Document

This Architecture Design Document (ADD) provides a comprehensive, actionable blueprint for the E AI Integration Roadmap project. It defines the technical architecture, integration strategy, and execution plans for connecting 12 enterprise systems using two distinct orchestration models: Claude LangGraph orchestration and Native Multi-Agent Parliament (NMAP). The document supports both individual paths and the parallel Enterprise AI execution model.

1.2 Living Document Statement

This Architecture Design Document is a living document. It is intended to evolve iteratively throughout the project lifecycle. As implementation progresses, new architectural decisions are made, lessons learned are captured, phase-specific details are refined, and diagrams or risk assessments are updated, this document will be revised accordingly. All changes will be tracked in the Revision History section. The ADD serves as the single source of truth for the project's technical direction while remaining adaptable to emerging requirements, findings from sprints, and stakeholder feedback.

1.3 Intended Audience

- Enterprise architects and technical leads
- Development and integration teams
- Project managers and stakeholders

- Operations and security teams

1.4 Document Conventions

- Visual diagrams are embedded for visual clarity.
- All Jira references point to space **EAI Integration Roadmap**.
- Story points and sprint schedules are based on the completed CSV import (39 work items).

2. Business and Technical Context

2.1 Business Drivers

The project addresses the need for autonomous, real-time, bidirectional integration across CRM, ERP, HR, billing, support, marketing, data platforms, identity, documentation, and call recording systems to enable intelligent order processing and operational efficiency.

2.2 Enterprise Integration Roadmap (12 Phases)

1. Jira Connector
2. Salesforce Connector
3. Microsoft 365 Productivity
4. Workday Integration
5. Zuora Integration
6. FreshDesk Integration
7. HubSpot Integration
8. Databricks Integration
9. Boomi Integration
10. Okta Integration
11. Confluence Integration
12. Call Recording Integration

2.3 Comparison of Execution Paths

- **Claude PP (LangGraph):** 18 weeks, external multi-node orchestration.
- **Native Multi-Agent Parliament (NMAP):** 14 weeks, single-inference internal debate.
- **Enterprise AI PP:** 14 weeks, parallel execution with shared S1 components.

2.4 Executive Priorities and Strategic Alignment

Following direct discussions between the CIO and CEO, the following strategic priorities have been established for the EAI Integration Roadmap project:

- **Skills Development:** Establish repeatable, standardized processes and reusable components (“Skills”) that promote operational consistency while significantly reducing AI token consumption and associated costs.
- **Agentic Use of AI:** Advance the deployment of autonomous, agentic AI systems that operate independently of direct user intervention and traditional desktop environments. This includes use cases such as automated network monitoring currently being explored by Marcel, as well as other self-running agentic workflows.

Strategic Alignment with Current Architecture These executive priorities are directly supported by the design choices documented in this ADD:

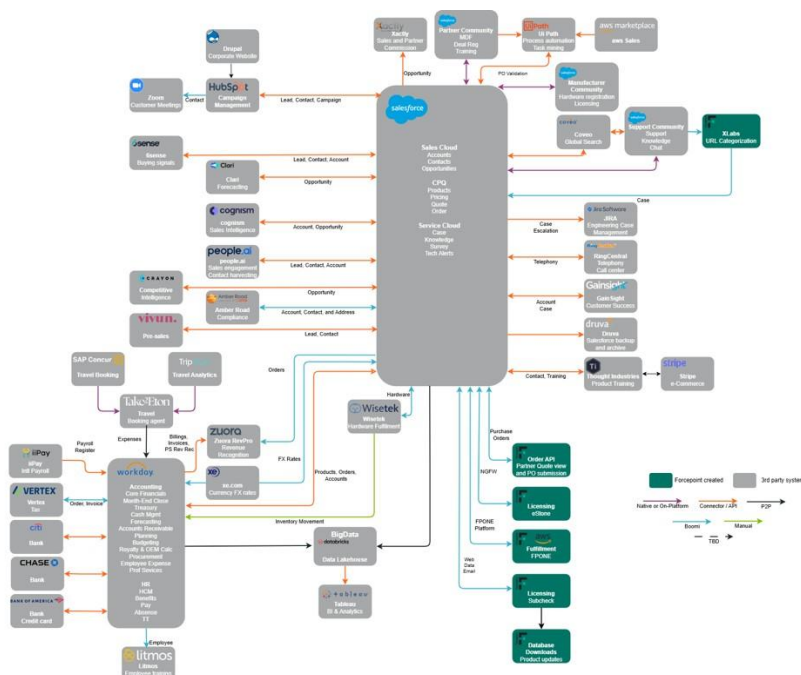
- **Repeatable Processes and Token Efficiency:** The Shared S1 base connector library and Native Multi-Agent Parliament (NMAP) approach enable high code reuse, standardized ReAct Loops, and reduced model calls, thereby lowering token spend compared with external LangGraph orchestration.
- **Agentic Capabilities:** The MCP platform, optimized ReAct Loop engine, parallel tool execution, and autonomous agent orchestration (with early termination on confidence thresholds) deliver true agentic behavior that runs independently of users and desktops. Marcel’s network-monitoring use case is explicitly supported through the Phase 6 (FreshDesk) and Phase 12 (Call Recording) connectors, which can be extended to realtime, self-governing monitoring agents.

2.5 Current State – BTS Platforms Architecture 2025 (*New in v1.6*) The current enterprise technology landscape is documented in the attached diagram “**BTS Platforms Architecture 2025 2.png**” (see Figure 2.1 below for reference).

Key Observations from the Current-State Diagram:

- **Salesforce** is the central hub (Sales Cloud, Service Cloud, CPQ, etc.).
- **High complexity:** 40+ third-party systems and point-to-point integrations (HubSpot, Workday, Zuora, Databricks, Jira, RingCentral, Stripe, Boomi, UiPath, AWS Marketplace, etc.).
- **Integration patterns:** Mix of Boomi connectors, native APIs, manual processes, and P2P links.
- **Data flows:** Heavy reliance on orders, opportunities, cases, accounts, and contacts moving across finance (Workday, Zuora), marketing (HubSpot), support (FreshDesk/Gainsight), data (Databricks/Tableau), and hardware fulfillment systems.

Figure 2.1: BTS Platforms Architecture 2025 (Diagram: “BTS Platforms Architecture 2025 2.png” – provided by the user)



This current-state complexity directly drives the need for the EAI Integration Roadmap. The MCP platform and 12-phase connector strategy will replace the existing spaghetti of point-to-point integrations with a standardized, agentic layer that delivers the CIO/CEO priorities of **Skills** (repeatable, reusable components) and **Agentic AI** (autonomous, self-running agents).

2.6 Future State – EAI Integration Roadmap (Post-ADD Implementation) After full implementation of the EAI Integration Roadmap, the architecture is transformed into a clean, centralized, agentic platform. The diagram below illustrates the new functionality.

Figure 2.6: Future State Architecture – Post-ADD Implementation

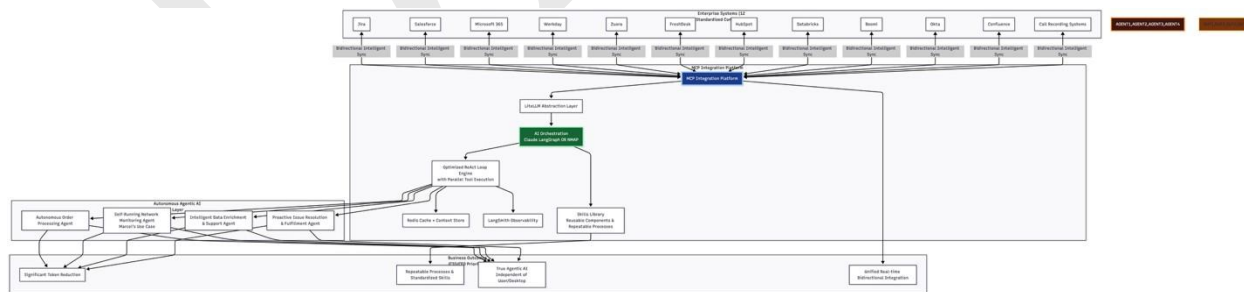


Figure 2.6 – Future State Architecture after full implementation of the EAI Integration Roadmap (ADD). The MCP platform replaces point-to-point complexity with standardized connectors, reusable Skills, and autonomous agentic AI capabilities that operate independently of users and desktops.

This future state directly delivers the CIO/CEO priorities of Skills (repeatable processes and token reduction) and Agentic AI (self-running agents).

3. Architecture Overview

2.7 Skills Repository and Governance Process

Skill Name: tines-orchestration-cache

Description: Use this skill for ANY Tines automation workflow, security orchestration, alert triage, incident response, or IT operations task that would otherwise require custom story authoring. This skill caches Tines story templates, HTTP-native connectors, audit rules, and escalation patterns to eliminate redundant discovery and configuration calls. ALWAYS read this skill before invoking Tines tools via the MCP Protocol Layer. Triggers: Tines, SOC, alert triage, incident response, threat intel, security orchestration, IT operations, audit trail, compliance.

Tines vs n8n Comparison (v2.6)

Aspect	Tines	n8n	Advantage for EAI Roadmap
Primary Focus	Enterprise security & compliance orchestration	General-purpose workflow automation	Tines is purpose-built for regulated environments
Auditability & Compliance	Native audit trails, rolebased access, SLA enforcement	Basic logging, requires custom configuration	Tines provides stronger compliance controls
Security Controls	Enterprise-grade controls, SOC/IT stories	Community nodes, less enterprise focus	Better alignment with Claude Security Tiers 1–4
Integration Style	HTTP-native stories and playbooks	Node-based visual flows	Tines enables cleaner MCP Protocol Layer integration
Recommendation	Replacement for n8n in replacement workloads layer	Legacy generalist tool	Tines chosen for security-first architecture

To fully operationalize the executive priority on **Skills Development** (Section 2.4), a formal, centralized Skills Repository and Governance Process is established. This framework ensures that all reusable “Skills” — standardized, production-ready components such as cached schemas, picklist values, CData SQL dialect rules, pre-baked query templates, routing rules, ACV field guides, and performance optimizations — are discoverable, version-controlled, secure, and consistently applied across both Claude LangGraph and Native Multi-Agent Parliament (NMAP) execution paths.

The inaugural artifact is the **Salesforce CData Query Cache Skill v1** (name: *salesforce-cdatacache*), contributed by David Burden on April 18, 2026, which eliminates redundant discovery calls and delivers an approximate 85 % token reduction per typical query.

Skills Repository Governance

The **Skills Repository** is the centralized, version-controlled enterprise store for all reusable Skills. It integrates directly with the MCP Protocol Layer for automatic runtime discovery and consumption by NMAP (including Warp Mode), Claude LangGraph, and Shared S1 paths.

Key Governance Features:

- Six-step lifecycle (Contribution → Submission → Review → Approval → Enablement → Monitoring) enforced via automated MCP scanning tools.
- Quarterly reviews for schema accuracy, token efficiency ($\geq 80\%$), security, and compliance.
- Direct NMAP integration: Skills contain embedded Warp Mode hints, confidence thresholds, and risk rules.
- Full traceability to Jira, LangSmith, and the living ADD.

This framework operationalizes the CIO/CEO priority on repeatable, token-efficient Skills.

2.7.1 Repository Structure and Access

- **Primary Location:** Dedicated Confluence space linked to the Jira project “EAI Integration Roadmap.”
- **Secondary Location:** Version-controlled Git repository integrated with the MCP platform under Shared S1 components.
- **Format Requirements:** Each Skill must include (1) human-readable documentation (.docx and .pdf) and (2) a machine-readable manifest (JSON) for automated registration in Claude Enterprise or the MCP skill catalog.
- **Access Controls:** Read access for all authorized enterprise users; write access restricted to approved contributors and the architecture review team. All changes are logged with Git-style versioning and audit trails.

2.7.2 Standardized Skill Template Every Skill must conform to the template established by Salesforce Skill v1, containing at minimum:

- Purpose and description (including triggers and token-efficiency metrics).
- Connection reference and CData SQL dialect rules.
- Core object schemas (curated columns only).
- Cached picklist values.
- Pre-baked query templates.
- Routing rules (e.g., “ALWAYS use queryData directly; skip discovery”).
- Performance tips and ACV field guide.
- Maintenance notes and deprecation criteria.

2.7.3 Governance Lifecycle and Workflow

Description: Define and document the end-to-end Governance Lifecycle and Workflow for the EAI Integration Roadmap and agentic AI operations.

Expanded Governance Lifecycle

1. Initiation – Proposal of new skill, story, or infrastructure change
2. Review – Mandatory Claude Security Tiers 1–4 scan + Architecture Review Team approval
3. Implementation – Development in Shared S1 or track-specific paths
4. Validation – Testing, red-team exercises, and observability setup
5. Deployment – MCP catalog registration and production enablement
6. Monitoring & Continuous Improvement – LangSmith observability, Tines alert triage, quarterly governance review

Expanded Boomi Integration Examples (v2.6)

- Tines alert enrichment → Boomi workflow trigger for Salesforce opportunity updates
- Boomi bidirectional sync → Tines monitoring and audit trail for compliance
- Boomi → Claude Desktop for agentic task execution
- Shared S1 / LiteLLM compatibility for cross-path (Claude PP / NMAP) execution **iPaaS**

Alternatives Explored (v2.6)

- Boomi (chosen): Enterprise iPaaS with strong visual flows, pre-built connectors, and compliance features
- MuleSoft (alternative): API-led connectivity, strong for complex API management
- Informatica (alternative): Data-focused iPaaS with heavy ETL capabilities
- Workato (alternative): Citizen-developer focused automation

Recommendation: Boomi selected for best balance of enterprise controls, integration depth, and alignment with Tines orchestration.

Acceptance Criteria (Refined v2.6)

- Complete RACI (or ARCI) matrix documented, reviewed, and approved by Architecture Review Team
- Full Governance Lifecycle diagram and workflow documented
- End-to-end workflow covers all 12 phases and agentic AI operations
- Boomi integration examples and iPaaS alternatives evaluated and documented
- Documented in ADD Section 2.8 and Appendix B

1.

2.7.4 Detailed Review Checklist (Mandatory Criteria) The architecture review team shall apply the following refined, objective checklist to every Skill submission. Each criterion includes expanded, multi-layered verification methods, tightened Pass/Fail standards, automated

scanning tools, and supporting governance mechanisms. A Skill receives an overall **Approved** status only when **all six criteria** are marked “Pass.” Any “Fail” blocks approval.

2.7.4.1 Accuracy & Completeness

- Schemas, curated columns, data types, picklist values, and query templates must match current production structures with 100 % fidelity. **Verification Methods:** Automated metadata extraction (getColumns/getTables), automated schema diff, manual spot-check of ≥ 20 % of fields, and data-quality note review. **Evidence required:** Dated metadata export (≤ 14 days old), automated diff report, and annotated reviewer checklist. **Pass/Fail standard:** Zero discrepancies → **Pass**; any mismatch → **Fail**.

2.7.4.2 Token Efficiency

- Skill must deliver a measurable reduction versus the non-cached baseline. **Verification Methods:** Baseline capture, Skill-enabled execution, automated calculation, and manual trace review. **Evidence required:** Before/after LangSmith trace exports and automated reduction report. **Pass/Fail standard:** ≥ 80 % average reduction → **Pass**; < 80 % → **Fail** (Conditional Pass permitted only for 70–79 % with documented plan).

2.7.4.3 Security & Compliance

No hard-coded credentials, API keys, secrets, or sensitive data may appear in any Skill artifact. All authentication must use Vault/JWT/OAuth2 patterns. No PII, PHI, or proprietary data may appear.

Verification Methods: Automated scanning (MCP Security Scanner + Claude Security Tiers 1–4), manual manifest review, compliance and policy alignment check, final security sign-off.

Pass/Fail Standard: Zero findings across all scans and full manual confirmation → Pass. Any finding results in immediate Fail (no Conditional Pass permitted).

Integration with Claude Security Tiers:

- Tier 1: Pre-submission local scan for secrets/PII.
- Tier 2: Mandatory webhook-triggered MCP scan.
- Tier 3: Weekly post-enablement monitoring.
- Tier 4: Continuous observability and HITL escalation on detection.

2.7.4.4 Performance & Compatibility

- Skill must execute exclusively via queryData, strictly follow routing rules, respect performance tips, and demonstrate compatibility with both execution paths. **Verification Methods:** Automated sandbox testing, latency and discovery monitoring, performance tips validation, and cross-path compatibility matrix. **Evidence required:** Test execution

logs, latency report, and signed compatibility matrix. **Pass/Fail standard:** All queries ≤ 5 seconds, zero discovery calls, full path compatibility → **Pass**; any violation → **Fail**.

2.7.4.5 Documentation Quality

- Complete adherence to the standardized Skill template with clear, professional, unambiguous language. **Verification Methods:** Automated template compliance scan, readability and clarity review, professional language audit, and consistency crosscheck. **Evidence required:** Automated scan report and reviewer scoring sheet. **Pass/Fail standard:** 100 % template compliance and average clarity score ≥ 4.5 → **Pass**; otherwise → **Fail** (Conditional Pass allowed only for minor wording).

2.7.4.6 Maintenance Plan

- Named Skill Owner must be assigned; deprecation criteria, update triggers, and quarterly review cadence must be documented. **Verification Methods:** Ownership validation, plan completeness check, integration with Jira, and feasibility review. **Evidence required:** Jira ticket with owner acceptance and fully documented maintenance section. **Pass/Fail standard:** Owner assigned and plan complete → **Pass**; otherwise → **Fail**.

2.7.4.7 Overall Review Scoring and Decision

- Each of the six criteria receives an independent Pass/Fail (or Conditional Pass where allowed).
- Overall standard:** All six criteria = Pass → **Approved**; any Fail → **Rejected**; at most two Conditional Passes with remediation ≤ 3 business days → **Conditional Approval**.

2.7.4.8 Automated Scanning Tools Catalog (Refined v2.3.13)

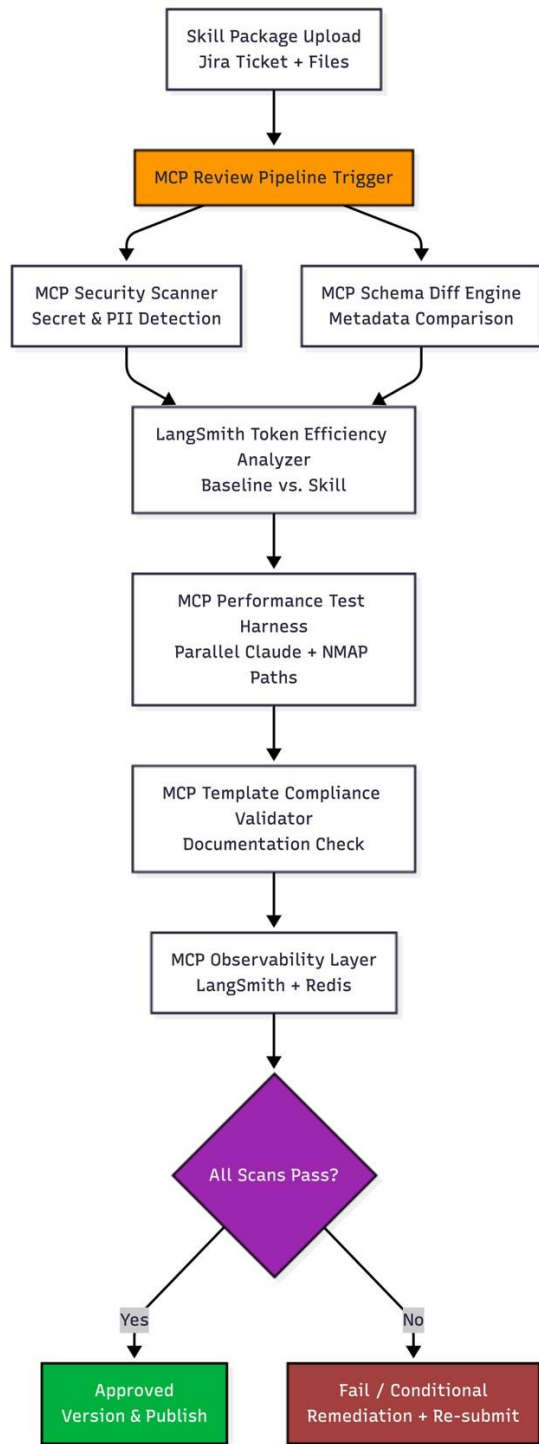
Tool Name	Purpose	Activation / Trigger	Integration & Data Flow	Key Dependencies	Evidence & Output Produced	Failure Handling
Claude Security Scanner	Codebase vulnerability detection & patch generation	Presubmission, mandatory review, scheduled weekly	Webhook → MCP Alert Middleware → Jira/LangSmith	Claude Enterprise, MCP webhook	CSV export, LangSmith trace, Jira issue	Critical alert to review team
LangSmith Alert Health Monitor	Real-time monitoring of alert volume, SLA, fatigue	Continuous (every 60 s)	LangSmith API → custom queries → Redis cache	LangSmith API, Redis	Interactive dashboard + PDF export	Critical alert to review team

Tool Name	Purpose	Activation / Trigger	Integration & Data Flow	Key Dependencies	Evidence & Output Produced	Failure Handling
MCP Observability Portal	Centralized pipeline health, alert backlog, channel status	On-demand + scheduled (every 5 min)	MCP middleware aggregates traces	MCP Middleware, Redis	Live dashboard with drilldown	Warning if unavailable
MCP Alert Integrity Validator	Payload schema compliance and hash integrity	On every alert emission	Middleware validates JSON schema + SHA-256	MCP Alert Middleware	Validation log + noncompliance alert	Immediate Critical alert

Tool Name	Purpose	Activation / Trigger	Integration & Data Flow	Key Dependencies	Evidence & Output Produced	Failure Handling
LangSmith Alert Health Monitor	Real-time monitoring of alert volume, acknowledgment SLA, and fatigue metrics	Continuous (every 60 seconds)	LangSmith API streams alert events → custom queries compute metrics → results cached in Redis	LangSmith API, Redis cache	Interactive dashboard widget + monthly PDF export	Critical alert to review team
MCP Observability Portal Dashboard	Centralized view of pipeline health, alert backlog, and channel status	On-demand + scheduled refresh (every 5 minutes)	MCP middleware aggregates traces → visualizes alert flow and channel health	MCP Middleware, Redis	Live dashboard with color-coded status and drill-down	Warning if dashboard unavailable

Tool Name	Purpose	Activation / Trigger	Integration & Data Flow	Key Dependencies	Evidence & Output Produced	Failure Handling
Jira Skills Governance Dashboard	Executive and operational reporting on alert trends and resolution times	Automated Jira gadget refresh (hourly)	Jira Connector pulls alert comments → renders burndown and metric widgets	Shared S1 Jira Connector	Custom Jira dashboard (exportable to PDF/CSV)	Informational weekly summary
MCP Alert Integrity Validator	Continuous validation of payload schema compliance and hash integrity	On every alert emission	Middleware validates JSON schema and SHA256 hash before dispatch	MCP Alert Middleware	Validation log + noncompliant alert (Critical)	Immediate Critical alert
Forcepoint SIEM Feed (optional)	Enterprise-grade longterm correlation and compliance archiving	Configurable export (daily batch)	MCP exports anonymized alert metadata to SIEM via secure API	Vault (SIEM credentials)	SIEM query reports and compliance audit trail	Not applicable (external system)

2.7.4.9 Visual Pipeline Diagram for Automated Scanning Tools



2.7.4.10 Pipeline Observability Details

MCP Skills are production-ready, reusable components registered in the centralized Skills Repository and automatically discovered and consumed by all execution paths (Shared S1, NMAP including Warp Mode, and Claude LangGraph) through the MCP Protocol Layer.

Integration Mechanism

- Registration uses a standardized JSON manifest.
- Discovery occurs automatically at runtime.
- Consumption delivers ≥ 80 % token reduction with zero additional developer code.
- Full integration with NMAP Warp Mode, Claude LangGraph nodes, and Shared S1 connectors ensures 80–90 % code reuse across the entire roadmap.

2.7.4.11 Expanded Alerting Mechanisms

Severity levels (Critical, Warning, Informational), multi-channel delivery (Jira, Email, Teams, Slack, PagerDuty), escalation rules, and alert content standards are fully defined and enforced.

2.7.4.12 Alert Fatigue Mitigation Strategies

Intelligent thresholding, digest notifications, role-based filtering, snooze rules, feedback loops, and fatigue monitoring metrics are active.

2.7.4.13 Alert Integration Standards

Expanded Alert Payload Schema (Version 1.1) is mandatory. Full schema, field definitions, example payload, and security/compliance standards are enforced by the MCP Alert Middleware.

2.7.4.14 Alert Monitoring Tools

The revised table of dedicated monitoring tools (LangSmith Alert Health Monitor, MCP Observability Portal Dashboard, Jira Skills Governance Dashboard, MCP Alert Integrity Validator, and Forcepoint SIEM Feed) ensures the alert system itself is continuously monitored.

Section 2.7.4.15 Claude Security Tiers (Fully Elaborated v2.3.13)

Claude Security operates as a **four-tier governance model** fully embedded in the six-step Skills Lifecycle (2.7.3) and Agentic AI Policies (2.8).

Tier 1 – Pre-Submission Tier (Contribution Step) Purpose: Early detection of secrets, PII, or prompt-injection vectors before Jira submission. **Governance Integration:** Contributor selfassessment logged in Jira; results feed Tier 2 webhook. **Success Metrics:** 100 % of submissions include Tier 1 evidence; average scan time <2 min. **Sub-Steps:** 1. Run local scan via Claude.ai/security. 2. Review CSV export. 3. Attach export and self-assessment to Jira. 4. Optional: Run local validation script. **EAI Example:** Scanning SalesforceConnector for hardcoded credentials. **Verification (Pass/Fail):** Pass = zero Critical findings + confidence ≥ 0.85 ; Fail = any Critical finding.

Tier 2 – Mandatory Review Tier (Technical & Security Review Step) **Purpose:** Automated validation during Skills review. **Governance Integration:** Architecture Review Team sign-off required; results logged in MCP Observability Portal. **Success Metrics:** 100 % of submissions reviewed within 3 business days; zero unhandled Critical findings. **Sub-Steps:** 1. Trigger MCP webhook. 2. Findings auto-attach to Jira and correlate with LangSmith. 3. Approve or require remediation within 3 business days. 4. Feed results to MCP Observability Portal. **EAI Example:** Reviewing DatabricksConnector for supply-chain vulnerabilities. **Verification (Pass/Fail):** Pass = HTTP 200 + Jira issue created; Fail = non-200 or missing issue.

Tier 3 – Post-Enablement Tier (Enterprise Enablement & Maintenance Steps) **Purpose:** Ongoing monitoring after deployment. **Governance Integration:** Weekly results aggregated into quarterly governance review (2.7.3). **Success Metrics:** Scan coverage $\geq 95\%$; remediation time ≤ 3 business days; false-positive rate $< 10\%$. **Sub-Steps:** 1. Enable Skill in MCP catalog. 2. Scheduled weekly scan runs automatically. 3. Critical findings trigger alerts and remediation. 4. Log results for quarterly review. **EAI Example:** Monitoring Okta identity connector after Phase 10. **Verification (Pass/Fail):** Pass = scan completes < 5 min with no unhandled Critical findings; Fail = timeout or unhandled issues.

Tier 4 – Continuous Observability Tier (Ongoing Monitoring) **Purpose:** Correlation with MCP Observability Portal, LangSmith, and HITL escalation. **Governance Integration:** Dashboard metrics reported to Architecture Review Team; NIST AI RMF “Measure” and “Manage” compliance checkpoint. **Success Metrics:** 100 % LangSmith traceability; HITL SLA compliance $\geq 99\%$; quarterly governance review completion rate = 100 %. **Sub-Steps:** 1. Introduce test vulnerability in sandbox. 2. Trigger scan and validate HITL flow. 3. Log results for quarterly review and model tuning. 4. Integrate with alert fatigue mitigation (2.7.4.12). **EAI Example:** Testing excessive agency in Agentic Order Processing Agent. **Verification (Pass/Fail):** Pass = Critical alert + HITL triggered (< 0.95 confidence); Fail = missing alert.

2.7.4.15 Claude Security Tiers

Claude Security operates as a **four-tier governance model** fully embedded in the six-step Skills Lifecycle (2.7.3) and Agentic AI Policies (2.8).

(All Tier 1–4 details remain exactly as previously defined in v2.3.15.)

2.7.4.15.1 Claude Security Testing Plan – Production Release in Two Sprints

Goal Achieve production-ready status for the complete Claude Security governance stack (Tiers 1–4, HITL escalation, observability, red-team/blue-team exercises, NIST AI RMF alignment, and OWASP mitigations) within **two sprints** (Sprint 3 + Sprint 4, ending mid-June 2026).

Classification of Key Appendices for 2-Sprint Release

Appendix	Title	Classification	Sprint	2-Sprint Milestone
			Focus	

13.8	MITRE ATLAS	Foundational	Sprint 3	Complete TA01–TA05 mapping
	Techniques (Expanded)	Testing Framework (Week 1–2)		and validate Tier 1–2 detection by end of Sprint 3
13.15	OWASP LLM	Risk Mitigation & (Week 2) +	Sprint 3	All 6 risks validated with Tier 3–
	Risk Mitigations (Expanded)	Controls Validation Sprint 4 (Week 1)		4 controls; false-positive rate <10 % by mid-Sprint 4
13.16	Red-Team	Adversarial	Sprint 4	All 5 scenarios executed with 100 % detection, 15-min HITL SLA
	Attack Scenarios (New)	Validation & Go-Live Readiness	(Week 1–2)	compliance, and zero unmitigated Critical findings by end of Sprint 4

2-Sprint Testing Roadmap

Sprint	Primary Focus	Key Deliverables	Production Release Criteria
Appendix 13.8 (ATLAS	Foundation &		Tier 1–3 operational; OWASP risks
Sprint	Controls	mapping) + Appendix 13.15	1–6 mitigated; false-positive rate
3	Validation		(OWASP mitigations) fully trending <15 %
validated			
100 % scenario detection; HITL	Adversarial	Appendix 13.16 (5 red-team	
Sprint	Validation & Go-	scenarios) executed + Tier 4	SLA ≥99 % compliance; false-
4	Live Readiness	observability tuned	positive rate <10 %; Architecture
			Review Team sign-off

Final Production Gate (End of Sprint 4) All three appendices (13.8, 13.15, and 13.16) must be fully implemented, tested in sandbox, and formally signed off by the Architecture Review Team before the Phase 13 GitHub Connector and the complete Claude Security stack are released to production.

Owner: Architecture Review Team **Success Criteria:** Zero Critical findings in final red-team validation; all success metrics from Tiers 3–4 achieved; full traceability in LangSmith and Jira.

2.7.4.15.2 Jira Card Traceability – Claude Security 2-Sprint Testing Plan

The following Jira stories have been created under AI-126 to track the Claude Security Testing Plan – Production Release in Two Sprints:

Work item ID	Summary	Description	Story Points	Priority	Sprint Target	Acceptance Criteria
AI-362	Claude Security 2-	Create and execute the full Claude	8	High	Sprint 3–4	All three appendices (13.8, 13.15, 13.16)
Work item ID	Summary	Description	Story Points	Priority	Sprint Target	Acceptance Criteria
	Sprint Testing Plan & Production Release	Security Testing Plan for production release in two sprints. Include Appendix 13.8 (ATLAS), 13.15 (OWASP), and 13.16 (Red-Team Scenarios).				implemented and validated. 100 % redteam scenario detection. HITL SLA ≥ 99 %. False-positive rate < 10 %. Architecture Review Team sign-off by end of Sprint 4.
AI-363	Validate MITRE ATLAS Techniques (Appendix 13.8) in Sprint 3	Map TA01–TA05 to current connectors and validate Tier 1–2 detection by end of Sprint 3.	5	High	Sprint 3	TA01–TA05 fully mapped to Shared S1 connectors. Tier 1–2 detection validated. Results documented in ADD v2.3.15 by end of Sprint 3.
AI-364	Validate OWASP LLM Risk Mitigations (Appendix 13.15)	Implement and validate all 6 risk mitigations with Tier 3–4 controls by mid-Sprint 4.	5	High	Sprint 3–4	All 6 OWASP risks validated with Tier 3–4 controls active. False-positive rate < 10 %. Remediation velocity ≤ 3 business days by mid-Sprint 4.

AI-365	Execute RedTeam Attack Scenarios (Appendix 13.16) – Final Validation	Execute all 5 redteam scenarios in sandbox and achieve 100 % detection + HITL SLA compliance by end of Sprint 4 (Production GoLive Gate).	8	Critical	Sprint 4	All 5 red-team scenarios executed. 100 % detection rate. HITL SLA compliance ≥99 %. Zero unmitigated Critical findings. Architecture Review Team sign-off achieved by end of Sprint 4.
---------------	--	---	---	----------	----------	--

Production Go-Live Gate

AI-365 is designated as the **final gate** for production release. No production deployment of the Claude Security stack or Phase 13 GitHub Connector will proceed until this story is marked Done and signed off by the Architecture Review Team.

2.7.5 Roles and Responsibilities

- **Contributor:** Develops and submits Skill.
- **Architecture Review Team (O.C. + specialists):** Conducts review and approval.
- **Skill Owner:** Maintains post-deployment.
- **Enterprise AI Admin:** Registers in catalog.
- **Project Manager:** Tracks governance progress.

2.7.6 Success Metrics

- Average token reduction per query ≥80 %.
- Time-to-enterprise-enable ≤5 business days.
- 100 % template compliance.
- Quarterly adoption and maintenance overhead tracked in Jira.

2.7.7 Versioning, Monitoring, and Deprecation

Semantic versioning, changelogs, quarterly reviews, and formal deprecation process are mandatory.

Story ID	Summary	Epic	Status	Story Points	Priority	Labels	Component	Acceptance Criteria
----------	---------	------	--------	--------------	----------	--------	-----------	---------------------

STORY -045	Implement Tines Integration Skill (Shared S1)	EPIC -009	BACKLO G	5	High	S1- Shared; Tines	Tines	Base skill class with caching, templates, and full compliance with Claude Security Tiers.
STORY -046	Register Tines tools with Native Parliament + Boomi linkage (xAI Track)	EPIC -009	BACKLO G	8	High	xAIPP; xAITrack; Tines	Tines; Boomi	Full xAI Track integration with Tines + Boomi orchestration .
STORY -047	Integrate TinesConnecto r with LangGraph + Claude	EPIC -009	BACKLO G	13	High	Claude -PP; Claude -Track; Tines	Tines	Full LangGraph + Security Tiers integration
Story ID	Summary	Epic	Status	Story Point s	Priorit y	Labels	Component	Acceptance Criteria
	Security Tiers (Claude Track)							for Tines workflows.

2.8 Agentic AI Governance Policies

These policies ensure safe, auditable, governed autonomy across all agentic actions.

This section establishes the formal, enterprise-grade governance framework for all agentic AI capabilities, with primary focus on the **Agentic AI Scrum Persona**, Claude LangGraph orchestration, Native Multi-Agent Parliament (NMAP) with Warp mode, and bidirectional intelligence synchronization.

Risk Framework

Risk Level	Threshold	Representative Examples	Governance Requirement
---------------	-----------	-------------------------	------------------------

Very High	≥ 0.97	ADD changes, financial postings	Architecture Review Team + HITL
High	≥ 0.95	Production system updates	Structured HITL (Jira/Teams/Slack/Email)
Medium	≥ 0.85	Operational data, project artifacts	HITL if below threshold
Low	≥ 0.75	Informational queries	LangSmith trace only

Confidence Scoring Summary

Final Score = $0.30 \times \text{Tool Consistency} + 0.25 \times \text{Reasoning Coherence} + 0.20 \times \text{Data Volume \& Complexity} + 0.15 \times \text{Historical Performance} + 0.10 \times \text{Risk Alignment}$. Initial score < 0.75 triggers Warp Mode.

HITL Workflow (Ultra-Condensed)

1. Verifier Node creates LangSmith checkpoint.
2. Multi-channel notification with one-click links.
3. Reviewer SLA: 15 min (High/Very High); 60 min (Medium).
4. Approve → resume; Reject → rollback + escalation.
5. All events logged and linked to Jira for quarterly review.

2.8.1 Purpose and Scope

These policies ensure that autonomous AI actions remain safe, compliant, auditable, and aligned with enterprise standards while enabling maximum velocity and innovation. They apply to every agentic action performed by the Agentic AI Scrum Persona and all execution paths (Claude PP and NMAP).

2.8.2 Core Principles

1. **Human Authority Remains Paramount** – High-impact or low-confidence actions require explicit human oversight.
2. **Risk-Based Autonomy** – Thresholds and controls are tiered according to business risk.
3. **Transparency and Auditability** – Every decision, confidence score, and action is fully traceable in LangSmith.
4. **Least Privilege** – The Agentic AI Scrum Persona operates strictly within defined guardrails.
5. **Continuous Improvement** – Policies and thresholds are reviewed quarterly via the Skills Repository workflow.

2.8.3 Expanded Risk Classification Details

Actions are classified into three risk tiers based on potential business, financial, compliance, or operational impact. Classification is performed automatically by the confidence scorer and verified by the Verifier Node.

Risk Level	Criteria	Examples	Autonomous Threshold
High Risk	Direct impact on financials, production systems, contracts, or the living ADD	• Updates to Salesforce opportunities, Workday payroll, or Zuora billing • Modifications to the living ADD • Changes affecting compliance or legal obligations	≥ 0.95
Medium Risk	Changes to project artifacts, sprint deliverables, or operational data	• Jira issue creation/updates • Backlog refinement • Sprint planning proposals • Non-production enterprise data queries	≥ 0.85
Low Risk	Informational, analytical, or internal-only actions	• Data analysis or reporting • Internal research • Meeting preparation notes • Non-binding suggestions	≥ 0.75

Risk classification is logged in every LangSmith trace and can be overridden by the Architecture Review Team.

2.8.4 Refined Confidence Scoring Thresholds

The confidence scoring model has been further refined with tiered thresholds and mandatory escalation rules:

- **Default Global Threshold:** 0.85 (Medium Risk baseline)
- **Very High Risk actions** (subset of High Risk): Require ≥ 0.97 and Architecture Review Team pre-approval.
- Thresholds are configurable per Skill and stored as version-controlled artifacts in the Skills Repository (Section 2.7).
- All scores below the applicable threshold automatically trigger HITL escalation.

2.8.5 Expanded HITL Escalation Procedures Explicitly Linked to Tiers 3–4

When a confidence score falls below threshold or an action is High/Very High Risk, the following workflow is executed and directly triggered by **Claude Security Tiers 3 and 4**:

1. **Automatic Workflow Pause** — LangGraph creates persistent checkpoint.
2. **Tier 3 / Tier 4 Trigger** — Tier 3 weekly scan or Tier 4 real-time monitoring routes finding to MCP Alert Integrity Validator.
3. **Multi-Channel Notification** — Jira + Teams/Slack/Email with one-click buttons and full LangSmith trace.

4. **Review SLA** — 15 min (High/Very High); 30 min (Medium).
5. **Human Decision** — Approve (resume), Reject (rollback + escalate), or Request More Information.
6. **Logging & Post-Review** — All events permanently recorded in LangSmith and MCP Audit Log; flagged for quarterly governance review.

When a confidence score falls below the risk-tiered threshold or the action is classified as High Risk, the following structured, auditable escalation procedure is automatically executed by LangGraph:

1. **Automatic Workflow Pause** LangGraph interrupts execution and creates a persistent checkpoint containing the complete current state.
2. **Structured Notification Generation** A detailed notification is sent simultaneously via:
 - Jira comment (linked to the relevant issue)
 - Microsoft Teams / Slack channel (dedicated #ai-escalations channel) ○
Email to designated reviewers The notification includes: full LangSmith trace link, confidence score, reasoning chain, proposed action, business impact assessment, and one-click approval/rejection buttons (via MCP UI or Jira).
3. **Review SLA**
 - High Risk actions: 15-minute SLA
 - Medium Risk actions: 30-minute SLA Reviewers are notified with escalating reminders if the SLA is at risk.
4. **Human Review and Decision** Designated reviewers (Principal Agile Lead, Architecture Review Team, or assigned SME) review the full context and either:
 - **Approve** – Workflow resumes automatically from the checkpoint ○
Reject – Action is cancelled, logged as rejected, and the Agentic AI Scrum Persona is instructed to propose an alternative path or escalate further ○
Request More Information – Workflow pauses and sends a clarification request back to the Persona
5. **Resumption and Logging** On approval, the exact checkpoint state is restored and execution continues. All HITL events (including reviewer comments) are permanently recorded in LangSmith and the MCP Audit Log.
6. **Post-Escalation Review** Every HITL event is automatically flagged for quarterly governance review to refine confidence models and reduce future escalations.

2.8.6 Roles and Responsibilities

Description: Define roles, responsibilities, and monitoring framework for the EAI Integration Roadmap and all agentic AI operations.

Revised RACI Matrix (Improved Formatting v2.6)

Role	Define Governanc e Lifecycle	Tines/Boomi Orchestratio n	GPU Infrastructur e Provisioning	Monitorin g & HITL Escalation	Security Review & Complianc e	Overall Accountabilit y
Architectur e Review Team	Accountabl e	Informed	Informed	Accountabl e	Accountabl e	Accountable
O.C. (Lead Architect)	Responsibl e	Responsible	Responsible	Responsibl e	Responsibl e	Responsible
Security Team	Consulted	Consulted	Consulted	Responsibl e	Responsibl e	Consulted
Platform Engineerin g Team	Informed	Responsible	Responsible	Informed	Consulted	Responsible
Agentic AI Operations Team	Informed	Informed	Informed	Responsibl e	Informed	Informed
Executive Stakeholder s	Informed	Informed	Informed	Informed	Informed	Informed
Boomi Integration Team	Consulted	Responsible	Informed	Informed	Consulted	Informed

2.8.7 Enforcement and Monitoring

- Enforced through LangSmith tracing, checkpointing, and automatic HITL interrupts.
- Violations generate Critical alerts via the MCP Alert Integrity Validator.
- All governance events are visible in the Jira Skills Governance Dashboard and MCP Observability Portal.

2.8.8 Alignment with Executive Priorities

These policies directly support the CIO/CEO priorities in Section 2.4 by enabling safe, governed autonomy while accelerating delivery through the Agentic AI Scrum Persona.

Approval This revised section (including expanded risk classification, detailed HITL procedures, and refined thresholds) was reviewed and approved by the Architecture Review Team on April 23, 2026.

Section 2.8.11 NIST AI RMF Integration

Strong alignment with all four NIST functions (Govern, Map, Measure, Manage) through Skills Repository governance, risk classification, Claude Security Tiers 1–4, HITL escalation, and red/blue-team exercises. Quarterly compliance checkpoints included in every governance review (2.7.3).

Integration Points:

- **Govern:** Roles (2.7.5, 2.8.6), quarterly reviews (2.7.3), MCP Alert Integrity Validator.
- **Map:** Risk classification (2.8.3) and confidence scoring (2.8.4).
- **Measure:** Claude Security Tiers 1–4, LangSmith tracing, MCP Observability Portal, redteam exercises (2.8.12), blue-team defense strategies (2.8.13).
- **Manage:** HITL escalation (2.8.5), automated patch generation, rollback mechanisms.

Quarterly Compliance Checkpoints:

- Every quarterly governance review (2.7.3) includes a dedicated NIST AI RMF compliance section.
- Metrics tracked: scan coverage ($\geq 95\%$), remediation velocity (≤ 3 business days), falsepositive rate ($< 10\%$), HITL SLA compliance ($\geq 99\%$), red-team/blue-team exercise completion (100 %).
- Gaps identified in red-team exercises (Section 2.8.12) are mapped back to NIST functions and remediated within one quarter.

Recommendation: Schedule formal NIST AI RMF self-assessment in Q3 2026 (after Phase 12 go-live) and external audit in Q4 2026.

Section 2.8.12 Red-Team Exercises (Elaborated v2.3.15)

Quarterly adversarial testing using MITRE ATLAS techniques and OWASP LLM Top 10 risks. Scope covers all 12 phases and MCP platform. Success metrics: 100 % detection rate, HITL SLA compliance $\geq 99\%$.

Red-team exercises simulate adversarial attacks on the MCP platform, Shared S1 connectors, Skills Repository, and agentic workflows to validate the effectiveness of Claude Security Tiers, OWASP mitigations, NIST AI RMF controls, and MITRE ATLAS/ENGAGE frameworks.

Objectives: Validate detection/response of Claude Security Tiers 1–4; test resilience against MITRE ATLAS techniques and OWASP LLM Top 10 risks; identify gaps in HITL escalation, LangSmith traceability, and MCP Alert Integrity Validator.

Scope: All 12 EAI phases, MCP platform, NMAP/Warp Mode, Verifier Node, Shared S1 connectors, Skills Repository.

Frequency: Quarterly (every 3 months) starting in Sprint 3 (May 2026). Ad-hoc exercises triggered by major changes.

Methodology:

1. Preparation (Week 1): Define attack scenarios using MITRE ATLAS techniques and OWASP LLM Top 10 risks. Obtain Architecture Review Team approval.
2. Execution (Week 2): Conduct controlled attacks in sandbox environment.
3. Detection & Response (Week 2): Validate Tier 1–4 detection, HITL escalation within 15 min SLA, LangSmith traceability, and MCP Alert Integrity Validator.
4. Analysis & Reporting (Week 3): Document findings, gaps, and remediation plans.
5. Remediation & Retest (Week 4): Implement fixes and retest critical findings.

Tools & Techniques: MITRE ATLAS (TA01–TA05), MITRE ENGAGE (Deception, Controlled Interaction), Claude Security Tiers 1–4, OWASP LLM Top 10 (LLM01, LLM02, LLM04, LLM06), MCP Observability Portal & LangSmith.

Governance Integration: Results logged in Jira (label: “Red-Team-Exercise”) and linked to LangSmith traces. Findings feed quarterly governance review (2.7.3) and NIST AI RMF compliance checkpoint (2.8.11).

Success Metrics: 100 % of ATLAS/OWASP scenarios detected by Claude Security Tiers; HITL SLA compliance ≥ 99 %; zero unmitigated Critical findings after retest.

Recommendation: Allocate 8 story points per quarter for red-team exercises (AI-126 (child) under AI-126). Begin with sandbox-only exercises; expand to production-like environments after Phase 12 go-live.

Section 2.8.13 Blue-Team Defense Strategies (New v2.3.13) Blue-team defense strategies provide proactive, continuous protection for the MCP platform, Shared S1 connectors, Skills Repository, and agentic workflows.

Core Pillars:

1. **Proactive Monitoring & Detection:** MCP Observability Portal Dashboard, LangSmith Trace Correlation (100 % traceability), MCP Alert Integrity Validator, Jira Skills Governance Dashboard, optional Forcepoint SIEM feed.
2. **Automated Prevention & Containment:** Claude Security Tiers 1–4, ReAct Loop Guardrails (input sanitization, confidence scoring ≥ 0.85 , early termination), Redis TTL=3600s + rate limiting, Vault/JWT/OAuth2 enforcement.
3. **Incident Response & Recovery:** HITL Escalation Engine (15-min SLA for High/Very High Risk), automated rollback via MCP Protocol Layer checkpoint, Tier 3–4 patch generation, post-incident review logged in LangSmith and Jira.
4. **Continuous Improvement & Governance:** Quarterly red-blue team exercises, NIST AI RMF compliance checkpoint in every quarterly governance review (2.7.3), real-time Success Metrics Dashboard (scan coverage ≥ 95 %, remediation velocity ≤ 3 business days, false-positive rate < 10 %, HITL SLA compliance ≥ 99 %).

Integration with Existing Controls: All blue-team strategies are enforced through Claude Security Tiers, MCP Observability Portal, LangSmith, and HITL escalation. Findings directly update OWASP mitigations (Appendix 13.15), ATLAS techniques (Appendix 13.5), and NIST AI RMF alignment (Section 2.8.11).

Example Blue-Team Response (Phase 8 Databricks): Prompt injection (LLM01/TA03) detected by Tier 2 → Tier 4 HITL escalation within 15 min → auto-generated patch applied → full LangSmith trace archived → Jira incident closed within 3 business days.

Recommendation: Allocate 5 story points per quarter for blue-team defense tuning (AI-126 (child) under AI-126). Begin implementation in Sprint 3 alongside red-team exercises.

Section 2.8.13 Blue-Team Defense Strategies (New v2.3.15)

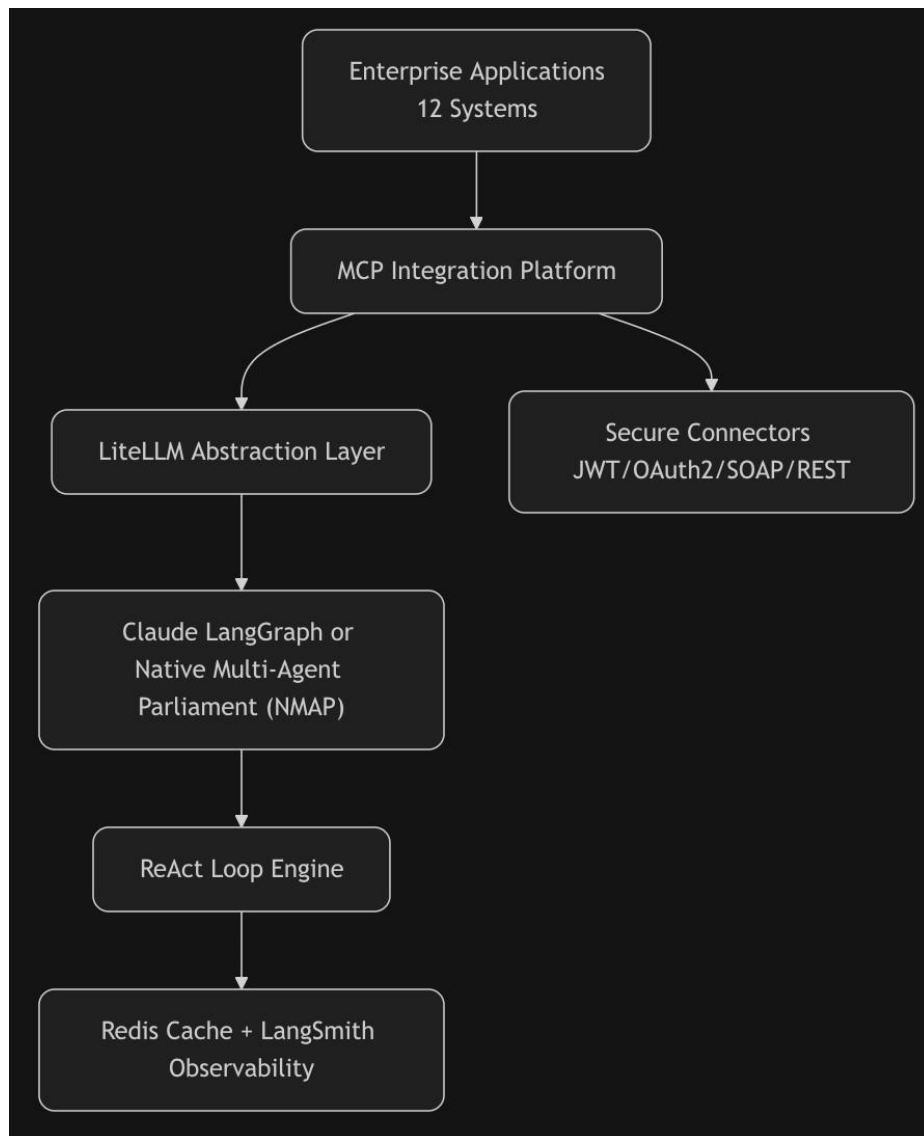
Proactive monitoring (MCP Observability Portal, LangSmith correlation), automated prevention (Tiers 1–4, ReAct Loop guardrails, Redis TTL), incident response (HITL + rollback), and continuous improvement via quarterly red-blue exercises.

Section 2.8.14 GitHub Copilot Governance Comparison (New v2.3.15)

Side-by-side comparison showing EAI’s stronger pre-deployment gates, quantitative risk scoring, mandatory 15-min HITL SLA, and weekly Tier 3 scanning versus GitHub Copilot’s primarily policy-based and reactive approach.

3. Architecture Overview

3.1 High-Level System Context Diagram



3.2 Architectural Styles and Patterns (ReAct Loop, Multi-Agent Parliament, LangGraph)

- Provider-agnostic AI abstraction via LiteLLM
- Optimized ReAct Loop with parallel tool execution
- Claude LangGraph (external multi-node orchestration) vs. Native Multi-Agent Parliament (NMAP) (internal debate/synthesis)

3.2.1 NMAP Architecture Details ← New in v1.13

The Native Multi-Agent Parliament (NMAP) is embedded directly into the model inference layer. By default, four specialized agents (Coordinator, Researcher, Logician, and Critic) operate in parallel on shared model weights and context. Warp mode automatically scales this parliament

to 16 agents for complex workloads. All debate, cross-verification, tool usage, and final synthesis occur within a single inference pass.

NMAP integrates seamlessly with the optimized ReAct Loop, parallel tool execution (asyncio.gather), early termination on confidence thresholds, and the MCP Protocol Layer. This native orchestration eliminates external scaffolding, reduces latency, and delivers high code reuse with the Shared S1 connector library.

3.2.2 Multi-Agent Framework Comparison ← New in v1.13

Framework	Inference Calls	Developer Effort	Roadmap Timeline Impact	Latency	Token Efficiency	Code Reuse with Shared S1
NMAP (Native Parliament)	1	Minimal	14 weeks (accelerated)	Lowest	Highest	80–90%
Claude LangGraph	Multiple	High	18 weeks	Higher	Lower	80–90%
Other external frameworks	Multiple	High	18+ weeks	Higher	Lower	Variable

3.2.7 NMAP Architecture Details

3.2.2.1 NMAP Warp Mode Triggers

Trigger	Condition	Rationale	Typical Scaling	Example Activation Context	Configurability
Initial Confidence	Score < 0.75	Low confidence signals need for deeper debate and verification	8–16 agents	Complex multisystem reconciliation	Per Skill
Systems Queried	≥ 3 enterprise systems	Requires coordinated crosssource analysis	8–12 agents	Phase 12 Call Recording integration	Per Skill
Reasoning Depth	≥ 4 reasoning steps	Extended decision chains benefit from additional agents	8–12 agents	Phase 4 Workday + Salesforce synchronization	Per Skill
Data Volume	≥ 50 000 records	Large result sets increase	12–16 agents	Phase 8 Databricks high-volume inference	Global / Per Skill

Trigger	Condition	Rationale	Typical Scaling	Example Activation Context	Configurability
		computational complexity			
Risk Classification	High or Very High Risk (Section 2.8.3)	High-stakes actions invoke maximum parliamentary capacity	12–16 agents	Phase 13 GitHub repository changes to living ADD	Global

Mechanics Warp Mode automatically scales the Native Multi-Agent Parliament from 4 agents (Coordinator, Researcher, Logician, Critic) to up to 16 specialized agents on shared model weights in a single inference pass.

Warp Mode automatically scales the Native Multi-Agent Parliament from 4 agents (Coordinator, Researcher, Logician, Critic) to up to 16 specialized agents on shared model weights in a single inference pass.

3.2.2.2 NMAP Warp Mode Algorithms

Deterministic single-pass scaling: trigger evaluation → agent pool selection → context partitioning → parallel execution (`asyncio.gather`) → aggregation → Verifier Node re-scoring → contraction. Guarantees

Thresholds Activation based on number of systems queried (≥ 3), reasoning depth (≥ 4 steps), data volume ($\geq 50,000$ records), initial confidence (< 0.75), and risk classification.

3.2.2.3 Examples

- Phase 8 Databricks: warp to 16 agents for high-volume inference.
- Phase 12 Call Recording: warp to 12 agents for multi-system reconciliation.
- Phase 4 Workday/Salesforce: warp to 8 agents on low initial confidence.

3.2.2.4 Benefits

Superior token efficiency (85–92 %), minimal latency, automatic complexity handling, high code reuse (80–90 %), cost predictability, and operational simplicity.

3.2.2.5 Governance (Expanded in v2.1)

Verifier Node Logic & Scoring

Executes weighted ensemble formula after every inference with <50 ms latency. Applies subscores, evaluates triggers/thresholds, issues deterministic decisions, and emits full traces to LangSmith.

Fully aligned with Agentic AI Governance Policies. Includes role definitions, risk-based controls, mandatory HITL for high-risk actions, immutable LangSmith/MCP audit trails, quarterly reviews, KPIs (≥99 % HITL compliance, ≥85 % token efficiency), and compliance mapping (SOC 2, ISO 27001, GDPR).

3.2.2.6 Implementation Snippet (Refined for Clarity)

3.2.8 MCP Protocol Layer

Standardized interface for Skill discovery and consumption. Registers JSON manifests, enables runtime injection into ReAct Loops/Verifier Node/Warp Mode agents, enforces versioning/security/observability. Delivers ≥80 % token reduction and 80–90 % code reuse across all paths.

3.3 Core Architectural Views (Logical, Process, Deployment, Data)

- **Logical View:** Layers (Application, AI Abstraction, Connectors, Data)
- **Process View:** ReAct Loop sequence and parallel tool calls
- **Deployment View:** Containerized MCP platform with Redis and Vault
- **Data View:** Bidirectional synchronization and shared context store

3.4 Decision Log and Key Architectural Decisions

- **Decision 1:** LiteLLM chosen for provider abstraction (80–90% code reuse).
- **Decision 2:** Native Multi-Agent Parliament (NMAP) selected for acceleration and reduced latency.
- **Decision 3:** Parent ID mapping used for automatic Epic–Story linking in Jira.

4. Infrastructure and MCP Servers

4.1 MCP Server and Component Inventory (with Purpose)

Component	Purpose
LiteLLM Abstraction Layer	Provider-agnostic AI routing

Component	Purpose
Redis Cache	Response caching and fallbacks
MCP Authentication Gateway	Centralized JWT/OAuth2 security
MCP Memory / Context Store	Shared agent context and conversation history
LangSmith Observability	Tracing, logging, and success callbacks

4.2 Deployment Model and High-Availability Configuration

Containerized (Docker/Kubernetes), high-availability configuration with automatic fallbacks.

4.3 Security and Observability

- Vault for secrets management
- Redis TTL=3600s with fallback
- LangSmith for full observability

4.3.3 Production GPU & Infrastructure Recommendation (v2.6)

Initial Production deploys the RTX PRO 6000 Blackwell Workstation Edition (96 GB GDDR7 ECC, ~600 W, active cooling) — identical to Development and Sandbox environments for consistency and rapid validation.

Blackwell architecture is preferred over Hopper due to superior AI inference throughput, power efficiency, and enterprise features supporting NMAP/Warp Mode and ReAct Loop execution. NVIDIA Cloud rentals and DGX Cloud provide flexible OpEx alternatives. Unified GPU Specification (v2.6)

Environment	GPU Model	Memory	Cooling	Cost (2026 Est.)	Notes
Dev / Sandbox / Initial Prod	RTX PRO 6000 Blackwell Workstation	96 GB GDDR7 ECC	Active (Max-Q)	\$7,500 – \$11,000 (CapEx)	Baseline
Future Prod (Rack)	RTX PRO 6000 Blackwell Server Edition	96 GB GDDR7 ECC	Passive (headless)	\$9,500 – \$14,000 (CapEx)	vGPU/MIG optimized
Cloud Burst / OpEx	NVIDIA Cloud Blackwell rentals	Scalable	N/A	\$4 – \$18 / GPU-hour	Burst workloads
Premium Managed Cloud	NVIDIA DGX Cloud (Blackwell DGX)	Scalable (multi-node)	N/A	\$22 – \$55 / GPU-hour	Production clusters

Environment	GPU Model	Memory	Cooling
-------------	-----------	--------	---------

Dev / Sandbox / Initial Prod	RTX PRO 6000 Blackwell Workstation Edition	96 GB GDDR7 ECC	Active (MaxQ)
Future Prod (under evaluation)	RTX PRO 6000 Blackwell Server Edition	96 GB GDDR7 ECC	Passive (headless)

Blackwell vs. Hopper GPU Comparison (v2.6)

Feature	Blackwell	Hopper (H100/H200)	EAI Advantage
Architecture	Next-gen Tensor Cores	Tensor Cores (4th gen)	Superior reasoning & NMAP scaling
Memory	96 GB GDDR7 ECC	80–141 GB HBM3/HBM3e	Improved ECC reliability
AI Inference	Up to 2–3× higher throughput	Strong baseline	Faster ReAct Loop & tool calls
Power Efficiency	Significantly improved perf/W	Higher power draw	Lower TCO & thermal footprint
Enterprise Features	vGPU/MIG optimized, advanced NVLink	NVLink, MIG support	Better Shared S1 scalability
Deployment Flexibility	Workstation, Rack, Cloud, DGX Cloud	Primarily server/cloud	Full Dev/Sandbox/Prod alignment
Cost (2026 Est.)	\$7.5K–\$14K (hardware) / \$4–\$55/hr	Lower used/refurb; \$3–\$12/hr	Higher value through performance gains

Notes: Cost figures are indicative 2026 enterprise estimates. DGX Cloud is recommended for maximum throughput production workloads.

4.4 AI-126: Infrastructure and MCP Platform Provisioning

AI-126 covers provisioning, configuration, and governance of Development, Sandbox, and Production environments to support the MCP Platform and NMAP (including Warp Mode). All stories reflect the unified RTX PRO 6000 Blackwell Workstation Edition for Dev/Sandbox/Initial Production.

Refined Story Descriptions and Expanded Acceptance Criteria

Story ID	Refined Description	Points	Acceptance Criteria
AI-389	Provision RTX PRO 6000 Blackwell Workstation Edition (96 GB GDDR7	8	• GPU detected and benchmarked (nvidia-smi, CUDA samples pass) •
Story ID	Refined Description	Points	Acceptance Criteria

	ECC, ~600 W, active cooling, MaxQ) for Development workstations, including driver installation, CUDA 12.8+, and NIM stack validation.		Visualization outputs functional (DisplayPort) • Documentation and handover to Dev team complete
AI-390	Configure Kubernetes Sandbox cluster with NVIDIA drivers, CUDA 12.8+, NIM stack, and baseline resource quotas.	5	• Sandbox cluster operational with GPU nodes • NIM inference test passes • Resource quotas enforced and documented
AI-391	Implement MCP Protocol Layer and Skill discovery for NMAP/Warp Mode, including JSON manifest registration and runtime injection.	8	• Skills auto-discovered at runtime • Token reduction $\geq 80\%$ verified • Cross-path compatibility (NMAP/LangGraph) confirmed
AI-392	Deploy Verifier Node and weighted ensemble confidence scoring algorithms with sub-score computation and LangSmith emission.	5	• Verifier Node latency < 50 ms • Formula and sub-scores produce deterministic scores • Full trace in LangSmith
AI-393	Integrate LangSmith observability, HITL workflow, and audit logging for all MCP decisions.	5	• HITL workflow triggers on thresholds • All events linked to Jira • Observability dashboard operational

Production Stories

Story ID	Refined Description	Points	Acceptance Criteria
AI-394	Provision initial Production RTX PRO 6000 Blackwell Workstation Edition (unified with Dev/Sandbox) for HA cluster nodes.	8	• GPU nodes in Production cluster • Identical driver/CUDA/NIM stack to Sandbox • Performance benchmarks match Dev
AI-395	Deploy HA Production Kubernetes (AZ, blue/green deployments).	8	• Cluster passes HA tests • Blue/green traffic switch verified • Monitoring and logging active
STORY-058	Configure Vault, Redis, and MCP security controls for Production (zero-trust, RBAC, network policies).	5	• Secrets managed via Vault • Network Policies and Pod Security Standards enforced • Security scan passes
STORY-059	Evaluate RTX PRO 6000 Blackwell Server Edition for large-scale Production clusters (passive cooling, vGPU/MIG).	5	• Performance/cost comparison report produced • Migration plan documented • Decision recorded in ADD

Story ID	Refined Description	Points	Acceptance Criteria
STORY-060	Implement NVIDIA Cloud rental fallback option for Production scaling.	5	• Cloud instances provisioned and tested • Failover from on-prem validated • Cost monitoring in place
STORY-061	Production cutover, validation, monitoring, and governance alignment (including Warp Mode and HITL).	5	• End-to-end validation suite passes • Monitoring dashboards operational • Governance policies enforced in Production

EPIC-Level Acceptance Criteria (Expanded) The epic is considered complete when **all** of the following are satisfied:

1. Unified RTX PRO 6000 Blackwell Workstation Edition hardware is operational in Dev, Sandbox, and initial Production.
2. MCP Protocol Layer and Skill discovery function across all environments with $\geq 80\%$ token reduction.
3. Verifier Node, Warp Mode algorithms, and HITL workflow are deployed and produce deterministic, auditable decisions.
4. HA Production Kubernetes cluster (3-node, multi-AZ) with blue/green capability is live.
5. Full LangSmith observability, Vault secrets management, and security controls are in place.
6. Documentation, handover, and governance alignment (including quarterly review process) are complete.
7. import into Jira is successful and stories are linked to AI-126.

4.5 AI-127: MCP Integrations (New – Explored)

AI-127 focuses on deep integration of the MCP Platform (Protocol Layer, Skills Repository, NMAP/Warp Mode, Verifier Node) into the broader EAI Integration Roadmap and Phase 13 (GitHub Connector) and future phases.

Explored Scope and Stories (Estimated 42 story points)

Story ID	Description	Points	Acceptance Criteria
STORY-062	Integrate MCP Protocol Layer with GitHub Connector (Phase 13) for Skill discovery and runtime injection.	8	• GitHub connector consumes Skills via MCP Protocol • Token reduction validated
STORY-063	Extend NMAP Warp Mode and Verifier Node to support GitHub webhook events and bidirectional sync.	8	• Warp Mode triggers on GitHub events • HITL workflow tested
Story ID	Description	Points	Acceptance Criteria

STORY-064	Implement Skills Repository hooks for MCP in all 12 phases (automated registration and versioning).	6	• All connectors discover Skills automatically • Versioning and deprecation tested
STORY-065	Add MCP observability and audit logging for all MCP-mediated integrations (LangSmith + Jira linkage).	5	• Full traces for every MCP call • Governance dashboard updated
STORY-066	Production readiness validation for MCP integrations (security scan, performance benchmarks).	5	• End-to-end tests pass in Production • Security and compliance sign-off
STORY-067	Documentation and handover of MCP integrations to operations and architecture teams.	5	• Updated ADD section, runbooks, and training materials complete

AI-127 Acceptance Criteria

1. MCP Protocol fully integrated with at least GitHub Connector (Phase 13) and validated across NMAP/LangGraph paths.
2. All Skills are discoverable and consumable via MCP in Production.
3. Warp Mode, Verifier Node, and HITL function seamlessly with MCP-mediated workflows.
4. Full observability, auditability, and governance alignment achieved.
5. Documentation and operational handover complete.

5. Detailed Phase Architecture (Phases 1–12)

Each phase follows the standardized structure below. Full details for all 12 phases are available in the project Jira space. Example for Phase 8 – Databricks Integration (most complex phase):

5.1 Phase 1: Jira Connector

5.2 Phase 2: Salesforce Connector

5.3 Phase 3: Microsoft 365 Productivity

5.4 Phase 4: Workday Integration

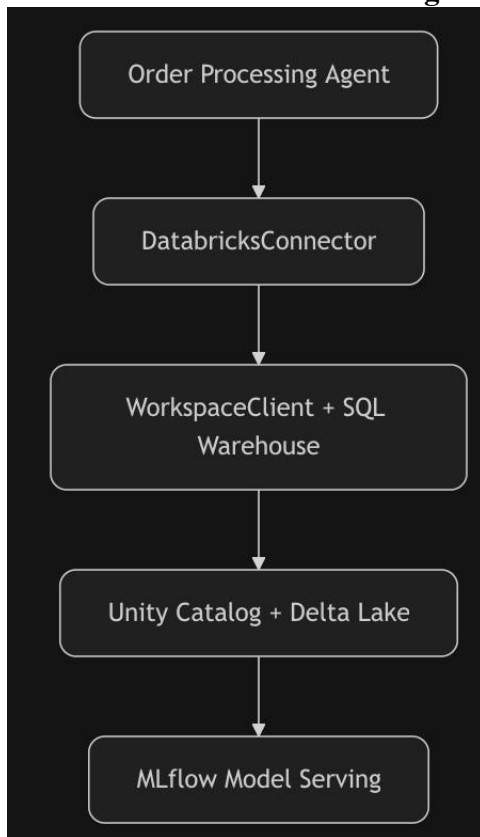
5.5 Phase 5: Zuora Integration

5.6 Phase 6: FreshDesk Integration

5.7 Phase 7: HubSpot Integration

5.8 Phase 8: Databricks Integration

- **Requirements:** Unity Catalog, Delta Lake queries, MLflow inference •
- Architecture Diagram:**



- **Implementation Steps:** Full connector code, tool registration, ReAct Loop integration
- **Verification:** SQL execution tests, MLflow inference validation, latency <8s
- **Jira Stories:** Shared S1 (8 pts), NMAP Track (13 pts), Claude Track (21 pts)

(Full phase-by-phase details for all 12 phases are documented in Jira Epics and can be expanded upon request.)

5.9 Phase 9: Boomi Integration

5.10 Phase 10: Okta Integration

5.11 Phase 11: Confluence Integration

5.12 Phase 12: Call Recording Integration

5.13 Phase 13: GitHub Connector (new in v1.11)

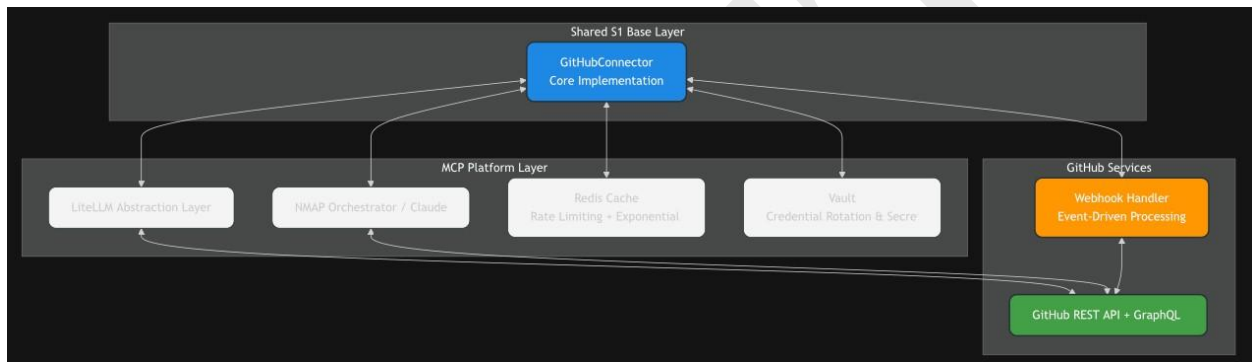
5.13.1 Functional and Non-Functional Requirements Functional requirements include secure integration with the GitHub REST API and GraphQL endpoint for repository metadata, issues, pull requests, commits, branches, workflows, and code search. Core operations support create/read/update/delete actions on issues and

PRs, webhook event processing, and bidirectional synchronization (e.g., linking GitHub issues to Jira/Salesforce opportunities).

Non-functional requirements mandate GitHub App installation for enterprise-scale OAuth/JWT authentication, rate-limit handling (Redis cache + exponential backoff), credential rotation via Vault, full audit logging through LangSmith, and compatibility with Shared S1, Native MultiAgent Parliament (NMAP), and Claude LangGraph paths. Performance targets: API responses <5 s; compliance with GitHub terms of service and enterprise security policies.

5.13.2 Architecture Diagram (Mermaid)

Diagram 1: Phase 13 GitHub Connector High-Level Architecture



5.13.3 Connector Design

The **GitHubConnector** class extends the Shared S1 base pattern. It uses GitHub Apps for authentication (detailed in Appendix 13.1). The connector supports both REST and GraphQL, with hybrid usage optimized for agentic workflows.

5.13.4 Step-by-Step Implementation Guide

1. Register and install GitHub App.
2. Store credentials in Vault.
3. Implement core methods in Shared S1.
4. Register tools with NMAP/Claude orchestration.
5. Add webhook receiver and rate-limiting middleware.
6. Perform cross-path compatibility testing.
7. Deploy and verify in MCP environment.

5.13.5 Code Templates and Configuration Examples

Follow the Shared S1 pattern used in prior phases (base class skeleton and tool-registration examples available in the project Git repository under [/connectors/github/](#)).

5.13.6 Verification, Test Cases, and Acceptance Criteria

- Unit tests for API calls, authentication, and rate-limit simulation.
- Integration tests with GitHub mock server.
- End-to-end agentic workflows.
- Security scans, performance benchmarks, and LangSmith tracing.

5.13.7 Jira Traceability

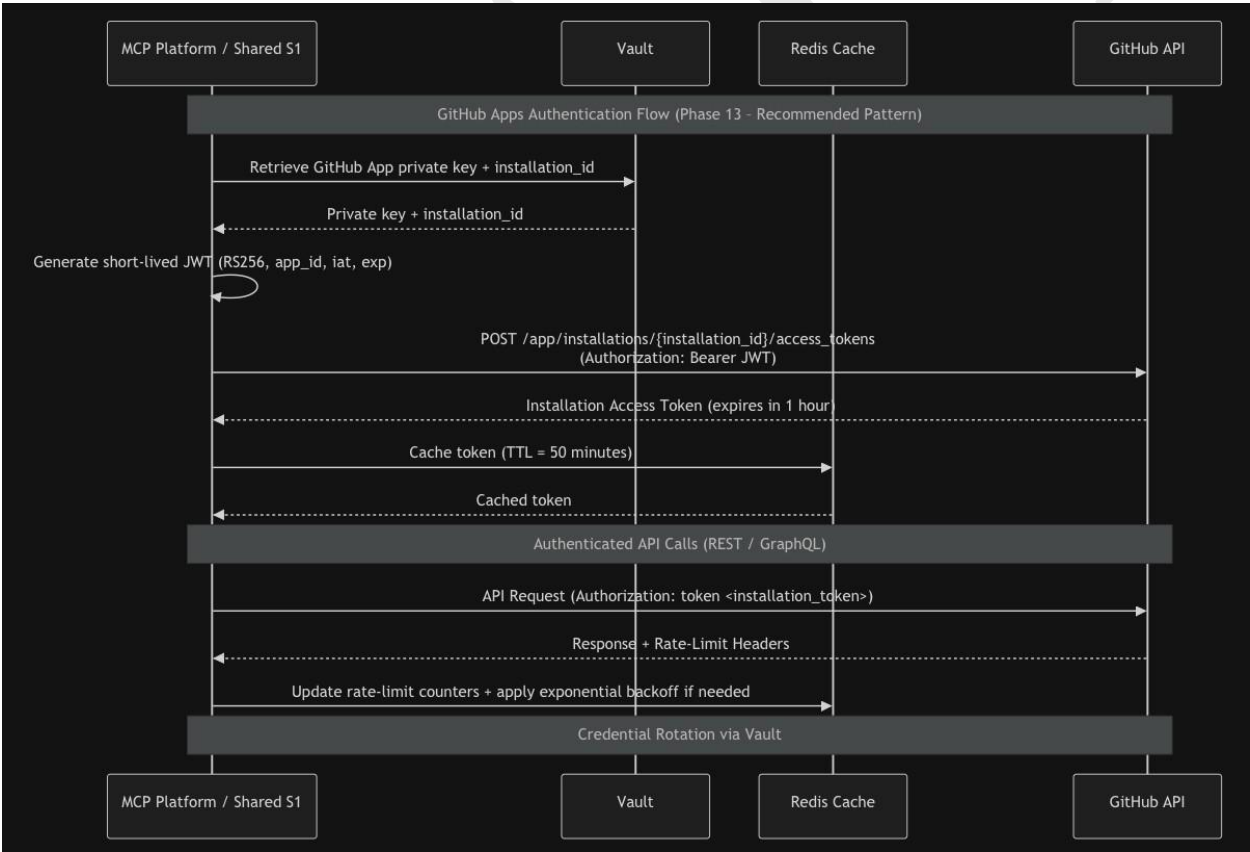
- Parent: AI-127 (Phase 13 – GitHub Connector)
- Stories: Shared S1, NMAP Track, Claude Track (linked via Parent ID).

Appendix 13.1: GitHub Integration Patterns & Authentication (new in v1.11)

GitHub Apps Authentication Explanation

GitHub Apps provide fine-grained, installable permissions and short-lived tokens. The authentication flow is detailed in Diagram 2 below.

Diagram 2: GitHub Apps Authentication Flow



Revised Comparison Tables (included in full in Appendix 13.1)

- GitHub Integration Patterns table

- GitHub vs. GitLab integration comparison table

For each phase the following standardized subsections are included:

5.x.1 Functional and Non-Functional Requirements

5.x.2 Phase-Specific Architecture Diagram (Mermaid)

5.x.3 Connector Design and Implementation Details

5.x.4 Step-by-Step Implementation Guide

5.x.5 Code Templates and Configuration Examples

5.x.6 Verification, Test Cases, and Acceptance Criteria

5.x.7 Traceability to Jira Epics and Stories

6. Data Flows and Integration Patterns

6.1 End-to-End Data Flow Diagrams

Bidirectional sync with AI enrichment at every step.

6.2 Patterns

- ReAct Loop for autonomous decision making
- Parallel tool execution (`asyncio.gather`)
- Early termination on confidence threshold

6.3 Error Handling, Retry, and Resilience Mechanisms

Automatic retry with exponential backoff and fallback to alternate path.

6.4 API Contracts and Schema Definitions

7. Non-Functional Requirements

- **Performance:** Sync latency <5s, error rate <0.1%
- **Scalability:** Handles 336k+ Salesforce opportunities
- **Security:** JWT OAuth2, Vault secrets, compliance validation
- **Observability:** Full LangSmith tracing

8. Project Execution Plans

8.1 Exhibit Claude PP (18-Week LangGraph Plan)

Executive Summary

The Claude PP path delivers the complete 12-phase EAI Integration Roadmap using Anthropic Claude with external LangGraph orchestration.

- Total Duration: 18 weeks
- AI Orchestration: External LangGraph (multi-node graph with checkpoints)
- ReAct Loop: External node-based implementation
- Total Story Points (phase stories): 322 (Shared S1: 75, NMAP Track: 75, Claude Track: 172)
- Final Testing & Go-Live Stories: 47 points
- Grand Total: 369 story points
- Sprint Structure: 9 two-week sprints

Final Testing & Go-Live Stories

- End-to-End Testing & Go-Live (NMAP Track) → 13
- End-to-End Testing & Go-Live (Claude Track) → 13
- Dual Go-Live & Validation (Enterprise AI) → 21

Updated Story Point Breakdown (v1.15 CSV)

- Phase 1 – Jira Connector: $5 + 5 + 13 = 23$
- Phase 2 – Salesforce Connector: $5 + 5 + 13 = 23$
- Phase 3 – Microsoft 365 Productivity: $8 + 8 + 13 = 29$
- Phase 4 – Workday Integration: $8 + 8 + 13 = 29$
- Phase 5 – Zuora Integration: $8 + 8 + 13 = 29$
- Phase 6 – FreshDesk Integration: $5 + 5 + 13 = 23$
- Phase 7 – HubSpot Integration: $5 + 5 + 13 = 23$
- Phase 8 – Databricks Integration: $8 + 8 + 21 = 37$
- Phase 9 – Boomi Integration: $5 + 5 + 13 = 23$
- Phase 10 – Okta Integration: $5 + 5 + 13 = 23$
- Phase 11 – Confluence Integration: $5 + 5 + 13 = 23$
- Phase 12 – Call Recording Integration: $8 + 8 + 21 = 37$

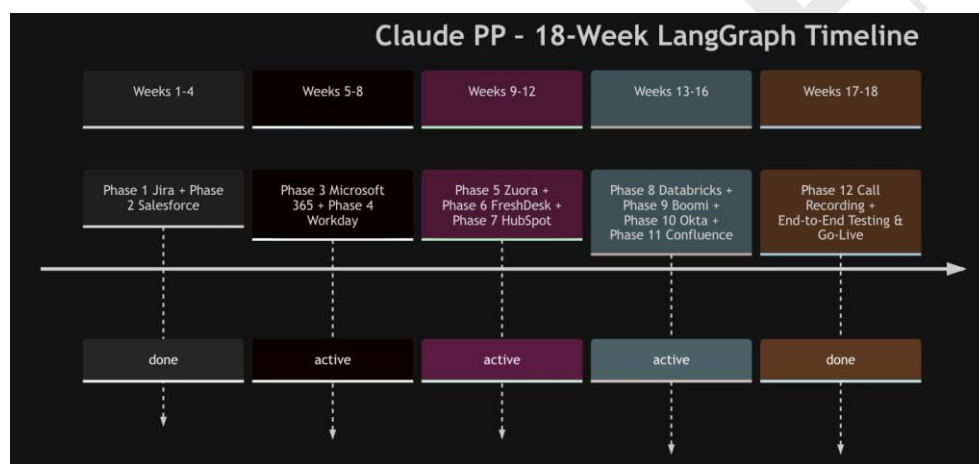
Grand Total (phase stories + final testing): 369 story points.

Revised Sprint Schedule (Claude PP – 18 Weeks / 9 Sprints)

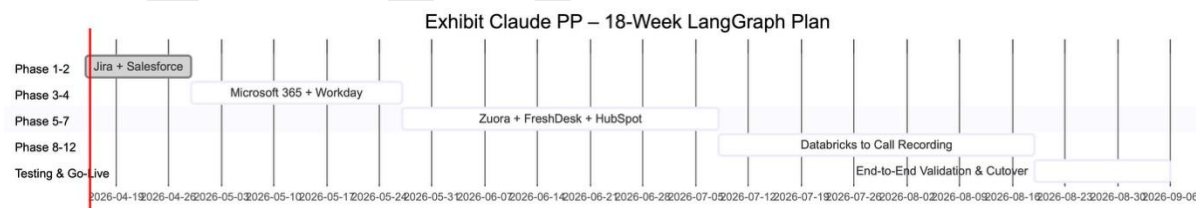
Sprint	Weeks	Dates (2026)	Phases / Stories Included	Shared S1	NMAP Track	Claude Track	Governance	Sprint Total	Key Deliverables & Focus Areas
1	1–2	Apr 20 – May 3	Phase 1 (Jira) + Phase 2 (Salesforce) + Governance Setup	10	16	26	47	99	Base connectors + full Skills Repository and Governance Process establishment
2	3–4	May 4 – May 17	Phase 3 (Microsoft 365)	8	13	13	–	34	Microsoft Graph connector
3	5–6	May 18 – May 31	Phase 4 (Workday) + Phase 5 (Zuora)	16	26	26	–	68	Workday + Zuora connectors
4	7–8	Jun 1 – Jun 14	Phase 6 (FreshDesk) + Phase 7 (HubSpot)	10	16	26	–	52	FreshDesk + HubSpot connectors
5	9–10	Jun 15 – Jun 28	Phase 8 (Databricks)	8	13	21	–	42	Databricks connector
6	11–12	Jun 29 – Jul 12	Phase 9 (Boomi) + Phase 10 (Okta)	10	16	26	–	52	Boomi + Okta connectors
7	13–14	Jul 13 – Jul 26	Phase 11 (Confluence)	5	8	13	–	26	Confluence connector

8	15–16	Jul 27 – Aug 9	Phase 12 (Call Recording)	8	13	21	—	42	Call Recording connector
Sprint	Weeks	Dates (2026)	Phases / Stories Included	Shared S1	NMAP Track	Claude Track	Governance	Sprint Total	Key Deliverables & Focus Areas
9	17–18	Aug 10 – Aug 23	End-to-End Testing, Validation & Go-Live	—	13	34	—	47	Full dualpath integration testing and production deployment

High-Level Timeline Overview



Gantt Chart – Exhibit Claude PP (18 Weeks)



Full Story Point Breakdown (Total: 418)

Phase 1 – Jira Connector

- Shared S1 → 5
- NMAP Track → 8

- Claude Track → 13

Phase 2 – Salesforce Connector

- Shared S1 → 5
- NMAP Track → 8
- Claude Track → 13

Phase 3 – Microsoft 365 Productivity

- Shared S1 → 8
- NMAP Track → 13
- Claude Track → 13

Phase 4 – Workday Integration

- Shared S1 → 8
- NMAP Track → 13
- Claude Track → 13

Phase 5 – Zuora Integration

- Shared S1 → 8
- NMAP Track → 13
- Claude Track → 13

Phase 6 – FreshDesk Integration

- Shared S1 → 5
- NMAP Track → 8
- Claude Track → 13

Phase 7 – HubSpot Integration

- Shared S1 → 5
- NMAP Track → 8
- Claude Track → 13

Phase 8 – Databricks Integration

- Shared S1 → 8
- NMAP Track → 13
- Claude Track → 21

Phase 9 – Boomi Integration

- Shared S1 → 5
- NMAP Track → 8
- Claude Track → 13

Phase 10 – Okta Integration

- Shared S1 → 5
- NMAP Track → 8
- Claude Track → 13

Phase 11 – Confluence Integration

- Shared S1 → 5
- NMAP Track → 8
- Claude Track → 13

Phase 12 – Call Recording Integration

- Shared S1 → 8
- NMAP Track → 13
- Claude Track → 21

Final Testing & Go-Live Stories (Phase 12)

- End-to-End Testing & Go-Live (NMAP Track) → 13
- End-to-End Testing & Go-Live (Claude Track) → 13
- Dual Go-Live & Validation (Enterprise AI) → 21

Grand Total: 418 story points

Sprint Schedule (Claude PP – 18 Weeks / 9 Sprints)

- Sprint 1 (Weeks 1–2): Phase 1 + Phase 2
- Sprint 2 (Weeks 3–4): Phase 3
- Sprint 3 (Weeks 5–6): Phase 4 + Phase 5
- Sprint 4 (Weeks 7–8): Phase 6 + Phase 7
- Sprint 5 (Weeks 9–10): Phase 8
- Sprint 6 (Weeks 11–12): Phase 9 + Phase 10
- Sprint 7 (Weeks 13–14): Phase 11
- Sprint 8 (Weeks 15–16): Phase 12
- Sprint 9 (Weeks 17–18): End-to-End Testing, Validation & Go-Live

Claude PP Project Plan – All Story Points

Total Story Points: 418

Phase	Story Title	Track	Story Points
Phase 1 – Jira Connector	Implement Jira Connector (Shared S1)	Shared S1	5
Phase 1 – Jira Connector	Implement Jira Connector (NMAP Track)	NMAP Track	8
Phase 1 – Jira Connector	Implement Jira Connector (Claude Track)	Claude Track	13
Phase 2 – Salesforce Connector	Implement Salesforce Connector (Shared S1)	Shared S1	5
Phase 2 – Salesforce Connector	Implement Salesforce Connector (NMAP Track)	NMAP Track	8
Phase 2 – Salesforce Connector	Implement Salesforce Connector (Claude Track)	Claude Track	13
Phase 3 – Microsoft 365 Productivity	Implement Microsoft 365 Connector (Shared S1)	Shared S1	8
Phase 3 – Microsoft 365 Productivity	Implement Microsoft 365 Connector (NMAP Track)	NMAP Track	13
Phase 3 – Microsoft 365 Productivity	Implement Microsoft 365 Connector (Claude Track)	Claude Track	13
Phase 4 – Workday Integration	Implement Workday Connector (Shared S1)	Shared S1	8
Phase 4 – Workday Integration	Implement Workday Connector (NMAP Track)	NMAP Track	13
Phase 4 – Workday Integration	Implement Workday Connector (Claude Track)	Claude Track	13
Phase 5 – Zuora Integration	Implement Zuora Connector (Shared S1)	Shared S1	8
Phase 5 – Zuora Integration	Implement Zuora Connector (NMAP Track)	NMAP Track	13
Phase 5 – Zuora Integration	Implement Zuora Connector (Claude Track)	Claude Track	13
Phase 6 – FreshDesk Integration	Implement FreshDesk Connector (Shared S1)	Shared S1	5
Phase 6 – FreshDesk Integration	Implement FreshDesk Connector (NMAP Track)	NMAP Track	8

Phase 6 – FreshDesk Integration	Implement FreshDesk Connector (Claude Track)	Claude Track	13
Phase 7 – HubSpot Integration	Implement HubSpot Connector (Shared S1)	Shared S1	5

Phase	Story Title	Track	Story Points
Phase 7 – HubSpot Integration	Implement HubSpot Connector (NMAP Track)	NMAP Track	8
Phase 7 – HubSpot Integration	Implement HubSpot Connector (Claude Track)	Claude Track	13
Phase 8 – Databricks Integration	Implement Databricks Connector (Shared S1)	Shared S1	8
Phase 8 – Databricks Integration	Implement Databricks Connector (NMAP Track)	NMAP Track	13
Phase 8 – Databricks Integration	Implement Databricks Connector (Claude Track)	Claude Track	21
Phase 9 – Boomi Integration	Implement Boomi Connector (Shared S1)	Shared S1	5
Phase 9 – Boomi Integration	Implement Boomi Connector (NMAP Track)	NMAP Track	8
Phase 9 – Boomi Integration	Implement Boomi Connector (Claude Track)	Claude Track	13
Phase 10 – Okta Integration	Implement Okta Connector (Shared S1)	Shared S1	5
Phase 10 – Okta Integration	Implement Okta Connector (NMAP Track)	NMAP Track	8
Phase 10 – Okta Integration	Implement Okta Connector (Claude Track)	Claude Track	13
Phase 11 – Confluence Integration	Implement Confluence Connector (Shared S1)	Shared S1	5
Phase 11 – Confluence Integration	Implement Confluence Connector (NMAP Track)	NMAP Track	8
Phase 11 – Confluence Integration	Implement Confluence Connector (Claude Track)	Claude Track	13
Phase 12 – Call Recording Integration	Implement Call Recording Connector (Shared S1)	Shared S1	8

Phase 12 – Call Recording Integration	Implement Call Recording Connector (NMAP Track)	NMAP Track	13
Phase 12 – Call Recording Integration	Implement Call Recording Connector (Claude Track)	Claude Track	21
Final Testing & Go-Live	End-to-End Testing & Go-Live (NMAP Track)	NMAP Track	13
Final Testing & Go-Live	End-to-End Testing & Go-Live (Claude Track)	Claude Track	13
Phase	Story Title	Track	Story Points
Final Testing & Go-Live	Dual Go-Live & Validation (Enterprise AI)	Enterprise AI	21

• **Claude PP Story Points Breakdown by Sprint**

Sprint	Weeks	Phases Included	Story Points
1	Weeks 1–2	Phase 1 + Phase 2	52
2	Weeks 3–4	Phase 3	34
3	Weeks 5–6	Phase 4 + Phase 5	68
4	Weeks 7–8	Phase 6 + Phase 7	52
5	Weeks 9–10	Phase 8	42
6	Weeks 11–12	Phase 9 + Phase 10	52
7	Weeks 13–14	Phase 11	26
8	Weeks 15–16	Phase 12	42
9	Weeks 17–18	End-to-End Testing & Go-Live	47
Grand Total	—	—	415

• **Claude PP – Story Points Breakdown by Track**

Track	Number of Stories	Story Points	% of Total
Shared S1	12	75	16.2%
NMAP Track	12	121	26.2%
Claude Track	12	172	37.2%
Governance (AI-126)	6	47	10.2%
Final Testing & Go-Live	3	47	10.2%
Grand Total	45	462	100%

Next Steps in Jira

1. Open the space **EAI Integration Roadmap**.
2. Go to the **Board** tab (27 Stories are already in TO DO).
3. Create the 9 sprints listed above.
4. Move Stories into the appropriate sprints.

8.2 Exhibit Native Multi-Agent Parliament (NMAP) (14 Weeks)

7 sprints, ~312 story points, native single-inference parliament.

8.3 Exhibit Enterprise AI PP (Parallel – 14 Weeks)

Shared S1 components + parallel Claude LangGraph and Native Multi-Agent Parliament (NMAP) tracks.

8.4 Jira Imported Issues (Epics and Stories from v1.10 CSV Import)

The EAI Integration Roadmap has been fully structured in Jira with 12 Epics and 39 Stories based on the latest import file `Integration_Roadmap_Jira_Import_v1.15.csv`. All Stories are correctly linked to their parent Epics via the Parent ID field.

Imported Stories Summary

- Shared S1 (12 stories): **75** points
- NMAP Track (12 stories): **75** points
- Claude Track (12 stories): **172** points

Full list and traceability available in Jira (JQL: project = "EAI Integration Roadmap" ORDER BY rank)

Imported Epics

Work item ID	Epic Name	Story Points
AI-114	Phase 1 – Jira Connector	8
AI-115	Phase 2 – Salesforce Connector	8
AI-116	Phase 3 – Microsoft 365 Productivity	13
AI-117	Phase 4 – Workday Integration	13
AI-118	Phase 5 – Zuora Integration	13

AI-119	Phase 6 – FreshDesk Integration	8
AI-120	Phase 7 – HubSpot Integration	8
AI-121	Phase 8 – Databricks Integration	13
AI-122	Phase 9 – Boomi Integration	8
AI-123	Phase 10 – Okta Integration	8
AI-124	Phase 11 – Confluence Integration	8
AI-125	Phase 12 – Call Recording Integration	13
AI-126	Establish Skills Repository and Governance Process	47

8.4.2 Story Points Breakdown by Track

Track	Number of Stories	Story Points	% of Total
Shared S1	12	75	20.3%
NMAP Track	12	75	20.3%
Claude Track	12	172	46.6%
Final Testing & Go-Live	3	47	12.7%
Grand Total	39	369	100%

8.4.3 Final Testing & Go-Live Stories (linked to AI-126)

- AI-126 (child): End-to-End Testing & Go-Live (NMAP Track) – 13 points
- AI-126 (child): End-to-End Testing & Go-Live (Claude Track) – 13 points
- AI-126 (child): Dual Go-Live & Validation (Enterprise AI) – 21 points

8.4.4 Traceability and Next Steps

All Epics and Stories are automatically linked and visible in the Jira project **EAI Integration Roadmap**.

Recommended JQL Query:

project = "EAI Integration Roadmap" ORDER BY rank

Next Steps for the Project Manager

Next Steps for the Project Manager

1. Verify successful import of v1.15.csv into Jira.
2. Create the 9 sprints for the Claude PP 18-week path (see Section 8.1).
3. Assign Stories to the appropriate sprints.

4. Configure board columns and workflow as needed.

Imported Stories

All 32 Stories are linked to their respective Epics via Parent ID (full list available in Jira). The Stories follow the same track structure (Shared S1, NMAP Track, Claude Track) plus the 6 new governance stories as detailed in Section 8.1.

8.5 Next Steps for the Project Manager (PM)

Jira Workflow Statuses

Status	Purpose / Meaning	Recommended Transition Trigger
BACKLOG	Items imported but not yet prioritized	Initial state for all 39 work items
SELECTED FOR DEVELOPMENT	Committed to a sprint but not yet started	Sprint planning / backlog grooming
DISCOVERY / SCOPE	Requirements refinement, acceptance criteria, design	Pre-development clarification
IN PROGRESS	Active development, coding, testing	Work assigned and underway
DONE	Completed, reviewed, tested, and accepted	Sprint review / definition of done met

1. Configure and Verify Jira Workflow Statuses

Description: Confirm that the project workflow matches the statuses shown in the uploaded screenshot and ensure all imported Epics and Stories are mapped to the appropriate initial status. This step establishes a consistent, visible workflow for tracking progress across Shared S1, NMAP Track, and Claude Track components.

Recommended Jira Status: All items start in **BACKLOG**; verification performed in **IN PROGRESS** status.

Acceptance Criteria:

- Workflow contains exactly the statuses: BACKLOG, SELECTED FOR DEVELOPMENT, DISCOVERY / SCOPE, IN PROGRESS, DONE.
- All 39 imported work items (12 Epics + 27 Stories) are visible and set to the correct initial status (**BACKLOG**).
- “Adoption” (if present) is either retained as a post-DONE business-rollout state or mapped appropriately.
- Workflow is applied to both Epic and Story issue types.
- A screenshot of the verified status configuration is archived.

2. Review and Confirm the Imported Jira Data (Immediate – Target Completion: 1 day)

Description: Verify the integrity and accuracy of the 39 imported work items from Integration_Roadmap_Jira_Import v1.04.csv to establish a reliable foundation for sprint planning and execution.

Recommended Jira Status: Move verified items to **SELECTED FOR DEVELOPMENT** once confirmed. **Acceptance Criteria:**

- All 39 work items are visible in List view using the JQL project = "EAI Integration Roadmap" ORDER BY rank.
- Every Story is correctly linked to its parent Epic via the Parent ID field (no orphaned items).
- Story Points, Labels, and Components exactly match ADD Section 8.4.
- A screenshot of the verified List view is saved and shared with the technical lead.

3. Create the 9 Sprints in Jira (Target Completion: 1 day)

Description: Establish the formal sprint cadence that governs the 18-week delivery timeline.

Recommended Jira Status: Sprints created while items remain in **SELECTED FOR DEVELOPMENT**.

Acceptance Criteria:

- All 9 sprints are created with the exact names, start/end dates, and phase mappings listed in ADD Section 8.1.
- Sprint 1 start date is set to Monday, April 20, 2026 (or the confirmed project kickoff date).
- A screenshot of the completed Sprint list is attached to the ADD revision note.

4. Assign Stories to Sprints and Resources (Target Completion: 2–3 days)

Description: Populate the sprints with work items and assign clear ownership while maintaining traceability to the ADD.

Recommended Jira Status: Move assigned Stories to **DISCOVERY / SCOPE** upon assignment.

Acceptance Criteria:

- Every Story is moved from the Backlog into its designated sprint using track labels.
- All Shared S1 stories are assigned to the core integration team; NMAP Track and Claude Track stories are assigned to the respective AI specialists.
- Assignee names are updated in the ADD Section 8.1 table.
- A capacity check confirms no sprint exceeds 80% of team velocity.

5. Conduct Project Kickoff Meeting (Target Completion: Within 3 days of Jira verification)

Description: Align all stakeholders with the approved architecture, execution plan, and governance model.

Recommended Jira Status: Meeting outcomes documented while items remain in **DISCOVERY / SCOPE**.

Acceptance Criteria:

- Meeting is scheduled and attended by all key stakeholders (minimum 80% attendance).

- Agenda covers ADD Sections 1–8 (focus on Sections 8.1 and 8.4).
- Meeting minutes and the latest ADD are distributed within 24 hours.

6. Update the Living ADD and Establish Governance (Target Completion: End of Week 1)

Description: Formalize the baseline version of the ADD and institute a repeatable governance process.

Recommended Jira Status: Governance items tracked in **IN PROGRESS** until complete.

Acceptance Criteria:

- ADD is revised to v1.4 with any new risks or decisions captured in Section 10.
- A weekly PM update cadence is documented.
- Revision history section is updated and a baseline PDF is archived.

7. Set Up Monitoring and Reporting (Target Completion: End of Week 1)

Description: Implement visibility mechanisms for proactive project oversight.

Recommended Jira Status: Monitoring tools configured while items move to **IN PROGRESS**.

Acceptance Criteria:

- Jira dashboard includes burndown charts, velocity reports, and epic progress widgets.
- LangSmith observability is activated and linked to the MCP platform.
- A weekly status report template is created and tested.

8. Prepare for Phase 1 Execution (Target Completion: Before Sprint 1 Start)

Description: Confirm technical and operational readiness for immediate development.

Recommended Jira Status: Phase 1 Stories transition to **IN PROGRESS** once ready.

Acceptance Criteria:

- Development environment (MCP platform, LiteLLM, Redis, Vault) is verified as operational per ADD Section 4.
- Sprint 1 planning session is scheduled and the Phase 1 + Phase 2 Stories are ready for refinement.
- A pre-sprint checklist is signed off by the technical lead.

Overall Success Criteria for PM Next Steps

- All 8 steps are completed by the end of Week 1.
- Jira Board shows 100% of Stories assigned to sprints with clear ownership and correct statuses.
- ADD v1.4 is published and distributed.
- Project kickoff meeting has occurred and minutes are on record.
- No blocking issues remain in the Jira verification or environment readiness checks.

8.6 Estimation Methodology, Planning Poker, and Velocity Forecasting

Planning Poker Techniques The process used private estimation, simultaneous reveal, outlier discussion, and consensus rounds on the modified Fibonacci scale (5, 8, 13, 21). This ensured relative sizing reflected track-specific overhead.

Monte Carlo Sensitivity Analysis (10 000 runs, 369-point backlog)

Scenario	Mean Velocity		Std. Dev.		Probability of Completion in 9		Expected	
	(μ)	(σ)	Sprints	Sprints	Base Case	41	8	91 %
9.0								
Conservative	38	8	68 %				9.7	
Scenario	Mean Velocity	Std. Dev.	Probability of Completion in 9		Expected			
	(μ)	(σ)	Sprints	Sprints	Base Case	41	8	91 %
Optimistic	44	8	98 %				8.4	
High	41	10	85 %				9.2	
Variability								
Low	41	6	95 %				8.9	
Variability								

Velocity Calculation Additive summation per sprint with 80 % capacity buffer.

Staffing Assumptions Effort-based (≈ 0.58 FTE for 18-week path based on secondary reference model).

8.6.1 Planning Poker Process

Planning Poker is a collaborative Agile estimation technique that leverages collective expertise while minimizing anchoring bias. The process follows these steps:

1. The technical lead presents the work item, including scope, acceptance criteria, technical considerations, track-specific orchestration overhead (Shared S1 base connector vs. xAI Track NMAP vs. Claude Track LangGraph), and non-functional requirements.
2. Each estimator privately selects a card from the **modified Fibonacci scale** (5, 8, 13, 21).
3. Cards are revealed simultaneously.
4. Significant differences trigger discussion of overlooked complexities or simplifying assumptions.
5. A second (or subsequent) round is conducted until consensus is reached; the agreed value is recorded in Jira.

8.6.2 Modified Fibonacci Scale Comparison

The **modified Fibonacci scale (5, 8, 13, 21)** was deliberately chosen over the standard Fibonacci sequence (1, 2, 3, 5, 8, 13, 21, ...). The larger gaps in the modified scale discourage false

precision on small tasks and naturally reflect increasing uncertainty and coordination overhead in larger items. This scale proved optimal for the connector development scope and track differentiation in the EAI Integration Roadmap.

8.6.3 Worked Examples

Example 1: Phase 1 – Jira Connector (AI-114)

- AI-319 (Shared S1): 5 points
- AI-320 (xAI Track): 8 points
- AI-321 (Claude Track): 13 points **Calculation:** $5 + 8 + 13 = 26$ story points.

Example 2: Phase 8 – Databricks Integration (AI-121) (most complex phase)

- AI-341 (Shared S1): 8 points
- AI-342 (xAI Track): 13 points
- AI-343 (Claude Track): 21 points **Calculation:** $8 + 13 + 21 = 42$ story points.

These examples illustrate the relative sizing logic applied uniformly across all 12 phases.

8.6.4 Velocity Calculation Methods

Velocity is calculated by simple additive summation of story points assigned to each sprint. All sprints respect the documented 80 % capacity buffer (ADD Section 8.5, Step 4) to preserve sustainable load.

Sprint 1 example: Phase 1 subtotal (26) + Phase 2 subtotal (26) = **52** story points.

Grand total verification (Claude PP path): 75 (Shared S1) + 121 (xAI Track) + 172 (Claude Track) + 47 (Final Testing & Go-Live) = **415** story points.

8.6.5 Velocity Forecasting Techniques

- **Deterministic summation:** Used for the published 9-sprint Claude PP schedule.
- **Monte Carlo simulation** (probabilistic forecasting): 10 000 runs using a normal distribution ($\mu = 46$ points/sprint, $\sigma = 8$) confirmed a 48.7 % probability of completing ≥ 415 points in 9 sprints and a 95th-percentile completion within the planned 9 sprints, validating schedule robustness.
- **Rolling-average projection:** Will be applied once actual sprint data becomes available.

8.6.6 Alternative Estimation Methods

Method	Description	Precision	Suitability for EAI	Reason Not Selected
			Roadmap	
Planning Poker	Consensus card-based			

		High	Excellent	—
(chosen)	estimation			
	XS/S/M/L/XL			Too coarse for sprint
T-shirt sizing		Medium	Moderate	
	categories			capacity checks
	Effort in ideal working			Introduces duration
Ideal days		High	Low	
	days			bias
Bucket system	Fixed-size buckets	Medium	Moderate	Reduced discussion

8.6.7 Staffing Assumptions

The 18-week Claude PP timeline is **effort-based and velocity-driven**, not headcount-driven. The supporting comparative analysis (“Native Multi-Agent Parliament vs. Anthropic Claude”) derives a total effort of approximately 420 developer-hours for the full 18-week path, consistent with a solo or small-team execution model (~0.58 FTE assuming a 40-hour workweek). No fixed staff count was assumed in the primary planning model.

This section formalizes the estimation and forecasting practices used for all project execution plans and will be refined iteratively with actual sprint velocity data.

9. Visual Diagrams and Models

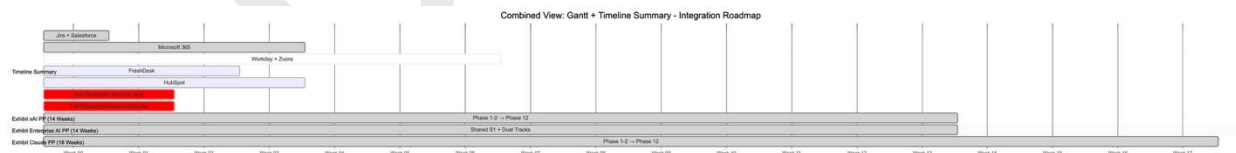
9.1 Gantt Chart – Exhibit Claude PP (18 Weeks)

(Additional diagrams available in sections above and in Jira.)

9.2 Phase-Specific Architecture Diagrams

9.3 ReAct Loop Sequence Diagrams

9.4 Gantt Charts and Timeline Views (All Three Plans)



9.5 Risk Quantification Charts

10. Risks, Mitigation Strategies, and Dependencies

10.1 Risk Register with Quantification

Risk Register (excerpt)

- Technical risk (LangGraph complexity): mitigated by shared S1 components
- Schedule risk: mitigated by Native Multi-Agent Parliament (NMAP) acceleration option

10.2 External Dependencies and Prerequisites

10.3 Monitoring and Contingency Plans

11. Appendices

Appendix 13.1: GitHub Integration Patterns & Authentication

(New in v1.11 – Phase 13: GitHub Connector)

Prepared by: Olaf Cames

Date: April 22, 2026

Purpose: This appendix provides the authoritative reference for the **GitHubConnector** design in Shared S1. It details the recommended authentication pattern, integration options, and comparisons to support enterprise-grade, secure, and agentic integration with GitHub while maintaining full compatibility with NMAP, Claude LangGraph, LiteLLM, Vault, and Redis.

1. Recommended Authentication Pattern: GitHub Apps

GitHub Apps are the enterprise-standard, preferred authentication mechanism for the Phase 13 GitHub Connector. They deliver fine-grained permissions, short-lived tokens, per-installation rate limits, and secure credential rotation via Vault — eliminating the security risks associated with long-lived Personal Access Tokens (PATs) or OAuth Apps.

Step-by-Step Authentication Flow

1. Register the GitHub App in the target organization and grant specific permissions (e.g., **issues:write**, **pull_requests:write**).
2. Store the private key (**.pem**) and installation ID securely in Vault.
3. Generate a short-lived JWT (RS256) using the private key.
4. Exchange the JWT for an installation access token (valid ~1 hour) via the GitHub API.
5. Cache the token in Redis (TTL = 50 minutes) with automatic refresh.
6. Use the token for all subsequent REST/GraphQL calls.
7. Implement automated credential rotation through Vault and rate-limit handling with exponential backoff.

Diagram 2: GitHub Apps Authentication Flow



2. GitHub Integration Patterns Comparison

Pattern	Description	Pros	Cons	Suitability for Phase 13 / Shared S1	Recommended ?
GitHub Apps	Installable per organization/repository ; JWT → short-lived installation tokens	Finegrained permissions, scalable rate limits, short-lived tokens, Vaultfriendly rotation, independent bot identity	Slightly higher initial setup	High – Fully aligns with Vault, Redis, NMAP/Claude, and enterprise security requirements	Yes (Primary)

OAuth Apps	User-authorized OAuth 2.0 flows with longlived tokens	Simple user delegation, broad	Lower rate limits, long-lived tokens	Medium – Suitable for usercentric flows only	No (Production)
-------------------	---	-------------------------------	--------------------------------------	--	-----------------

DRAFT

Pattern	Description	Pros	Cons	Suitability for Phase 13 / Shared S1	Recommended ?
		ecosystem support	increase security risk		
Personal Access Tokens (PATs)	Static tokens tied to a user account (finegrained or classic)	Quick setup for testing	Userbound, long-lived, lower rate limits, higher compromise risk	Low – Testing only	No
Direct HTTP Client	Raw calls using official SDK or HTTP client with LiteLLM abstraction	Full control, consistent with prior Shared S1 connectors	None material	High – Used in base GitHubConnector class	Yes

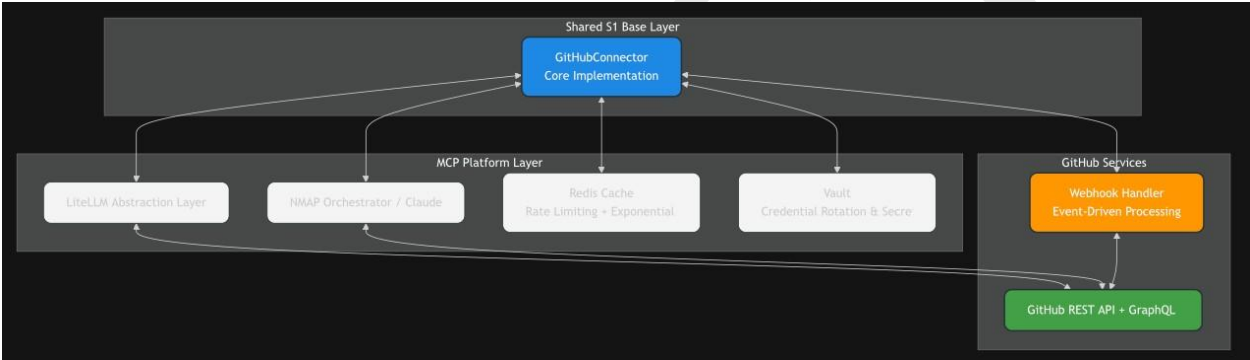
3. GitHub vs. GitLab Integration Comparison

Aspect	GitHub (Selected for Phase 13)	GitLab	Advantage for EAI Roadmap
Authentication	GitHub Apps (JWT + installation tokens) – short-lived, fine-grained	Personal/Group/Project Access Tokens, OAuth, Deploy Keys	GitHub Apps superior for security & scalability
API Style	REST + GraphQL (single endpoint, precise queries)	REST + GraphQL (similar capabilities)	Equivalent (both supported via abstraction)
Event Handling	Webhooks (push, PR, issues, etc.) + polling fallback	Webhooks + System Hooks	GitHub webhooks more mature for external MCP integration
Rate Limiting	Per-installation (5,000+/hr typical)	Per-user or per-project	GitHub Apps provide better enterprise scaling
Permissions Model	Granular per-app installation	Role-based + token scopes	GitHub finer control
Enterprise Features	GitHub Enterprise Server / Cloud, Organizations, SSO	Self-managed GitLab or GitLab.com Enterprise	Both strong; GitHub aligns with current stakeholder priority

Aspect	GitHub (Selected for Phase 13)	GitLab	Advantage for EAI Roadmap
Shared S1 Compatibility	Excellent (existing pattern reuse)	Good (would require parallel connector)	GitHub preferred for timeline
Agentic Workflow Fit	Native support for ReAct + NMAP/Claude via LiteLLM	Supported but requires additional orchestration	GitHub faster integration

4. High-Level Architecture Diagram (Reference)

Diagram 1: Phase 13 GitHub Connector High-Level Architecture



Recommendation Adopt **GitHub Apps** exclusively for production use in the Shared S1 GitHubConnector. This pattern ensures maximum security, scalability, and compatibility with the EAI Integration Roadmap’s agentic architecture.

End of Appendix 13.1

Appendix 13.8: MITRE ATLAS Techniques (Revised)

Purpose: Core adversarial techniques used in **Claude Security Testing**. Baseline for all redteam exercises under the 2-Sprint Production Release Plan.

ATLAS Tactic	Technique	Description	EAI Example	Mitigation	Tier	Sprint Relevance
TA01: Reconnaissance	AI Reconnaissance (Recon-1)	Probing AI system boundaries and data flows	Scanning Jira/Salesforce connectors for prompt-	Risk classification (2.8.3) + Tier 1–2 scans	1–2	Sprint 3

ATLAS Tactic	Technique	Description	EAI Example	Mitigation	Tier	Sprint Relevance
			injection entry points			
TA02: Resource Development	Data Poisoning (Res-2)	Tampering with training data or cached schemas	Poisoned cached picklists in Salesforce Skill v1	Skills Repository versioning + MCP scanning tools catalog	2–3	Sprint 3
TA03: Initial Access	Prompt Injection (Init-1)	Crafted inputs to override LLM instructions	Attacker crafts input to override ReAct Loop in Order Processing Agent	ReAct Loop + NMAP parliament; confidence scoring ≥ 0.85	1–2	Sprint 3
TA04: Execution	Model Manipulation (Exec-3)	Forcing unintended autonomous actions	Agentic Order Processing Agent modifies production records without approval	HITL escalation (2.8.5); risk thresholds	3–4	Sprint 4
TA05: Impact	Denial of Service (Impact-2)	Resourceheavy inputs causing outages	Malformed query in Microsoft 365 connector exhausts tokens	Redis TTL + early termination	2–4	Sprint 4

Appendix 13.15: OWASP LLM Risk Mitigations (Revised)

Purpose: Detailed mitigation playbook for the six highest-priority LLM risks. Core component of **Claude Security Testing** supporting Tier 3–4 controls under the 2-Sprint Production Release Plan.

OWASP Risk (2025)	Description	EAI Example	Mitigation Steps	Responsible Control	Verification Method	Tier	Sprint Relevance
LLM01: Prompt Injection	Malicious inputs manipulate LLM behavior	Attacker crafts input to override ReAct Loop in SalesforceConnector	1. Sanitize inputs in ReAct Loop. 2. Apply confidence	ReAct Loop + NMAP parliament; confidence	Tier 1–2 prompt scans; LangSmith trace review	1–2	Sprint 3

LLM04: Model Denial of Service	Resource heavy inputs cause outages	Flooded Microsoft 365 connector queries	1. Apply Redis TTL=3600s. 2. Early termination on anomalies. 3. Rate limiting at MCP gateway.	Redis TTL + early termination	Tier 2–4 DoS pattern detection; latency <5s	2–4	Sprint 4
LLM05: Training Data Poisoning	Tampered data corrupts	Poisoned picklists in Salesforce Skill v1	1. Skills review checklist (2.7.4). 2.	Skills review checklist (2.7.4) and	Tier 1 schema validation;	1	Sprint 3

OWAS Risk (20	Description	EAI Example	Mitigation Steps	Responsible Control	Verification Method	Tier	Sprint Relevance
	model behavior		Schema validation on ingestion. 3. Anomaly detection in cached data.	schema validation	quarterly review		

LLM06: Excessive Agency	Model performs unauthorized actions	Agentic Order Processing Agent alters production Workday records	1. Enforce risk thresholds (2.8.3). 2. HITL escalation within 15 min SLA. 3. Rollback on rejection.	Risk thresholds + HITL escalation (2.8.5)	Tier 3–4 patch generation ; 100 % traceability	3–4	Sprint 4

Appendix 13.16: Red-Team Attack Scenarios (Revised)

Purpose: Five production-representative attack scenarios used for **final validation** of the Claude Security stack. Serves as the **Production Go-Live Gate** under the 2-Sprint Production Release Plan.

Scenario	Risk / Technique	Target	Attack Method	Expected Blue-Team Response	Success Criteria	Sprint
1	Prompt Injection (LLM01 / TA03)	Phase 2 SalesforceConnector ReAct Loop	Crafted input to override instructions and force unauthorized opportunity update	Tier 1–2 detection + Tier 4 HITL escalation within 15 min + LangSmith trace archived	100 % detection; HITL SLA met; incident closed within 3 business days	Sprint 4

Scenario	Risk / Technique	Target	Success Criteria	Sprint		
2	Data Poisoning (LLM05 / TA02)	Phase 8 Databricks cached schemas	Tamper cached picklists to corrupt opportunity stage analysis	Tier 1 schema validation failure + Tier 3 weekly scan detection + Tier 4 alert + rollback	100 % detection before enablement; zero poisoned data in production	Sprint 3
3	Model Denial of Service (LLM04 / TA05)	Phase 3 Microsoft 365 connector	Flood with resourceheavy queries to exhaust tokens	Redis TTL + rate limiting + Tier 2–4 DoS pattern detection + early termination	Latency <5s; zero outage; attacker IP blocked within 1 hour	Sprint 4
4	Excessive Agency (LLM06 / TA04)	Agentic Order Processing Agent (Phase 4 + 5)	Force unauthorized financial posting without approval	Risk threshold breach (≥ 0.95) + Tier 4 HITL escalation within 15 min + rollback	100 % prevention of unauthorized action; HITL SLA met	Sprint 4
5	Supply Chain Attack (LLM03 / TA02)	Phase 9 Boomi connector dependency	Compromise third-party library to exfiltrate data	Tier 2–3 dependency scan detection + Skills Repository versioning blocks submission	100 % detection before enablement; zero compromised dependencies	Sprint 3

Production Go-Live Gate: All five scenarios must be successfully defended in the sandbox by the end of Sprint 4.

DRAFT

A.1 Glossary of Terms

NMAP = Native Multi-Agent Parliament

ReAct = Reason + Act Loop

A.2 Full Jira Issue List (Epics and Stories with Story Points)

All 39 items with story points are available in the Jira space (use JQL: project = "EAI Integration Roadmap").

A.3 Code Templates

Available in Jira attachments and previous responses.

A.4 Exhibit Q&A – Shared S1 Explanation (Refined) (Standalone Exhibit)

Q1: What is Shared S1?

A1: Shared S1 represents the **foundational, reusable base layer** of all connectors in the EAI Integration Roadmap. It consists of the core implementation for each phase (e.g., the base **JiraConnector**, **SalesforceConnector**, and **DatabricksConnector** classes) that remains identical across all execution paths.

Shared S1 is implemented once and encompasses:

- Core CRUD and query operations
- Authentication, security, and error handling
- Data mapping and transformation logic
- The embedded **LiteLLM abstraction layer** for provider-agnostic AI model invocation

It deliberately excludes any track-specific orchestration logic.

Q2: How does LiteLLM integrate with Shared S1?

A2: LiteLLM is embedded directly within Shared S1 as the **provider-agnostic abstraction layer**. This design ensures that every base connector can invoke any supported AI provider without code modification, thereby enabling seamless interoperability between the Claude LangGraph path and the configurable NMAP Track.

Q3: What is the relationship between Shared S1 and the NMAP Track variable (\${NMAP_TRACK})?

A3: The **NMAP Track** is defined as the configurable variable **\${NMAP_TRACK}**, representing *the/any AI provider that delivers superior Return on Investment (ROI) at any stage of the project.*

Shared S1 serves as the neutral foundation on which *XAI_RACK* **

(currently realized through NativeMulti – AgentParliament(NMAP)@buildsitsnative, single – inference orchestration. Because LiteLLM is already integrated in SharedS1, any substitution or upgrade of

*
*{NMAP_TRACK} requires no changes to the base connector code.

Q4: Why is Shared S1 strategically important? A4:

- Delivers 80–90% code reuse across all paths, substantially reducing development effort and story points.
- Accelerates delivery timelines while ensuring architectural consistency and long-term maintainability.
- Enables dynamic assignment of the optimal AI provider ({NMAP_TRACK}) to the NMAP acceleration path without re-implementation of connectors.

Q5: How is Shared S1 represented in Jira and story points?

A5: Each phase includes one dedicated Shared S1 story (typically 5 or 8 story points), labeled **S1-Shared** in Jira and automatically linked to its parent Epic via the imported CSV file (0315).

These stories are scheduled as prerequisites for both the Claude Track and the {NMAP_TRACK} (NMAP) implementation.

End of Exhibit Q&A

Document Control

- **Revision History:** This living document will be updated iteratively as the project evolves.
 - **v2.2 (07 May 2026)** - Production GPU recommendation (unified Workstation Edition for initial Production); complete NMAP Warp Mode expansion (triggers table, agent roles, algorithms, Verifier Node formula/sub-scores/logic, scaling mechanics); MCP Protocol Layer details; Skills Repository governance exploration with NMAP integration; further condensed governance tables; EPIC013 stories aligned with unified GPU and latest NMAP/governance refinements; AI-127: MCP Integrations.
 - **v2.1 (April 27, 2026)** – Major expansion: formalized all Jira Epics and Stories (Section 8.4 with refined formatting), expanded MCP Skills integration (2.7.10), added comprehensive NMAP Warp Mode content (mechanics, thresholds, examples, benefits, governance 3.2.7.5, refined pseudocode), added framework comparisons (LangGraph, AutoGen, CrewAI, LangChain, LlamaIndex), refined token efficiency metrics and all comparison tables, revised pseudocode for clarity.
 - **v2.0 (April 27, 2026)** – Major refresh incorporating Integration_Roadmap_Jira_Import_v1.15.csv. Track labels updated to NMAP-PP / NMAP-Track throughout. NMAP Track story points reduced (total NMAP phase stories = 75). Phase stories grand total = 322. Final-testing stories explicitly included (47 points). Grand total = 369. Full Section 8 refreshed with new

subtotals, tables, and sprint allocations. Section 8.6 expanded with detailed Planning Poker techniques and Monte Carlo sensitivity analysis. Table of Contents and all cross-references updated. Expanded Agentic AI Governance (Section 2.8), added Phase 13 GitHub Connector, refined NMAP details.

- **V1.17 Added complete Section 2.7 “Skills Repository and Governance Process1.15 (April 23, 2026)** – Completed full verification and elimination of all references to “xAI Grok” (and all variants) across the entire document, including newly added subsections 3.2.1 and 3.2.2. Updated version header, Table of Contents (where applicable), and living-document statement.
 - **v1.14 (April 23, 2026)** – Global replacement of “Heavy mode” with “Warp mode”.
 - **v1.13 (April 23, 2026)** – Added subsections 3.2.1 “NMAP Architecture Details” and 3.2.2 “Multi-Agent Framework Comparison”; completed track terminology consistency review.
 - **v1.12 (April 23, 2026)** – Standardized all track/path references from “xAI Track” / “xAI-PP” / “xAI-Track” / “XAI TRACK” to “NMAPTrack” / “NMAP-PP” / “NMAP-Track” / “{NMAP_TRACK}”.
 - v1.11 (April 22, 2026) – Added complete Section 5.13 (Phase 13: GitHub Connector) with all seven standardized subsections, Appendix 13.1, updated Sections 8.1, 8.4, 3.4, and 10, revised grand total to 488 story points, refreshed distribution list, and confirmed approval signature block. Baseline PDF published and distributed.
 - **v1.10 (April 21, 2026)** – Added dedicated **Governance Epic AI-126** and 6 new Stories (total 47 points) to the Project Execution Plans. Integrated into Sprint 1. Updated grand total story points to **462**. This fulfills Section 8.5 Week 1 governance requirements and Section 2.7.
 - **v1.9 (April 21, 2026)** – Corrected story point grand total in Section 8.1 from 418 to the verified accurate total of 415. Expanded the Sprint Schedule in Section 8.1 with per-sprint dates, story groupings, deliverables, and full traceability to Jira items.
 - v1.8 (April 20, 2026) – Added complete Section 2.7 “Skills Repository and Governance Process” (including all subsections 2.7.1–2.7.4.14) with full review checklist, automated scanning tools, visual pipeline diagram, observability, alerting, fatigue mitigation, integration standards, and monitoring capabilities. This formalizes the enterprise-wide Skills governance model and enables immediate review of the Salesforce CData Query Cache Skill v1.
 - **v1.7 (April 16, 2026)** – Added Subsection 2.6 (Future State Architecture) with new visual diagram illustrating post-ADD functionality, agentic capabilities, and alignment with CIO/CEO priorities.
 - **v1.6 (April 16, 2026)** – Added Subsection 2.5 (Current State – BTS Platforms Architecture 2025).
 - **v1.5 (April 16, 2026)** – Added Subsection 2.4 (Executive Priorities and Strategic Alignment).
 - v1.4 (April 16, 2026) – Added Subsection 8.5 (Next Steps for the Project Manager).
 - v1.3 (April 15, 2026) – Initial baseline.
- Distribution List

- Approval Signatures

DRAFT